

**THE
BROOKLYN MEDICAL JOURNAL.**

Vol. II. Brooklyn, N. Y., August, 1888. No. 2.

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Title: The Brooklyn Medical Journal. Vol. II. No. 2. Aug., 1888

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Release date: October 13, 2019 [eBook #60493]
Most recently updated: October 17, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/60493

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BROOKLYN MEDICAL JOURNAL. VOL. II. NO. 2. AUG., 1888 ***

**THE
BROOKLYN MEDICAL JOURNAL.**

PUBLISHED MONTHLY BY THE MEDICAL SOCIETY OF THE
COUNTY OF KINGS.

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VOL. II. BROOKLYN, N. Y., AUGUST, 1888. No. 2.

ORIGINAL ARTICLES.

PAIN, WITH SPECIAL REFERENCE TO ITS DENTAL RELATIONS.

BY WM. M. THALLON, M.D.

Read before the Brooklyn Dental Society, May 28, 1888.

MR. PRESIDENT AND GENTLEMEN:—Some months ago, when sitting in the operating-chair of your Chairman of the Committee on Subjects, he asked me if I would not read a paper before the Brooklyn Dental Society. In the helpless condition in which I then was, with literally a gag in my mouth, robbing me of the prerogative of free speech, and under the shadow of a formidable mallet, I somewhat timorously signified an assent. Under those circumstances I know of few men who would have had the moral and physical courage to have resisted such an appeal. When in the course of his further practices, he asked me what my subject would be, I promptly replied by mentioning the thing then most vivid in my mind: Facial Neuralgia.

I hardly realized my rashness and what I had undertaken, until I received your printed bulletin of subjects. But it has seemed to me on further thought that we might perhaps spend an hour profitably together in comparing notes about that borderland of facts and problems, which you touch on the one side as dentists and I on the other as physician. And I trust you will be lenient with me in your judgments if I go astray in my talk, and I pray you to remember that we doctors labor under great disadvantages compared with you dentists, contrasting the width and vagueness of our territory of research with the precision and accuracy of yours. I have again and again envied the exquisite dexterity and the certainty of adapting means to ends which I have seen exhibited by members of your profession, and vainly longed for the same in my own. But on the other hand, I think it may justly be urged that the dentists have not contributed as much to the general stock of knowledge, especially to the solution of disputed questions of pathology, such as the relation of micro-organisms to disease, as their unrivaled opportunities for observation would allow.

I shall therefore not hesitate, Mr. President, to somewhat dogmatically present my views on certain subjects, but I ask you to believe it is mainly

because I hope the gentlemen present will honor them by frank and full discussion.

I shall also ask permission to change the subject of my remarks from the announced title to one of a little wider scope, namely, Pain, with special reference to its dental relations.

I presume the symptom of pain is the one for which the overwhelming number of your patients, as the majority of ours, apply to us for relief. And yet common as this sensation is both in ourselves and in others, it is very remarkable how little settled opinion is, as to its nature. If you have never had occasion to try and put into the form of a definition the idea of pain, and proceed to consult the authorities, you will be surprised that so many different views could be held of what at first seems so common and obvious as to be beyond dispute. As you proceed in your inquiries, the question instead of becoming simpler apparently becomes more complex, for as you think of the different forms of pain, and contrast, for instance, that of an inflamed rheumatic joint, with its definite structural changes and well-marked constitutional symptoms like fever, with an idiopathic neuralgia, pure and simple, often lacking in any outward manifestation other than the pain itself, you wonder if the pains resulting are not as different as the diseases producing them. But the common consciousness of mankind which has given the same name to the sensation produced, whether by an inflamed bowel or a carious tooth, is sure to be right in believing that there is essentially the same substratum in each. Now what is the nature of that substratum? It is evident that whatever else it is, pain is a disagreeable sensation, and the word sensation further obliges us to remember that it involves a central nervous system (in its simplest type a single cell), capable of feeling impulses, conveyed to it from without, or else generated within itself. Now, it is very evident that pain must consist either in some change in the nature of the impulses sent to our central cell, or else in some change in the condition of the receiving centre. So eminent an authority as Prof. Erb defines pain simply as an increase in the ordinary sensory stimulus, a heightening more or less intense of ordinary sensation. On the other hand, Anstie defines pain as a perturbation in the nervous system, especially of the central cells, involving a lowering of function, a diminution of ordinary sensation. It is very evident that both of these great authorities cannot be exclusively right, and I propose to see what light we could get on this subject from the abundant clinical evidence you have.

This question is no mere quibble about words or definitions, but it is one of the utmost practical importance in its relation to treatment. According as we settle in our minds whether a given case of pain is an exaggeration or a lessening of the ordinary physiological condition, our treatment will logically be either narcotic or stimulant.

Leaving for the present the question as to the nature of pain, let us examine some of the modes in which it expresses itself; and as far as practicable I will limit myself to the various pains about the head, for all the varieties are there manifested.

The first point which strikes every observing man is the difference of individuals in their susceptibility to pain. It is not merely or even mainly a question of the amount of courage of the patient in bearing pain, but it is far more a question of inherited or acquired sensitiveness. The same amount of injury, as nearly as we can judge, in two differently organized individuals will produce extremely differing degrees of pain. In general it may be stated that the unduly susceptible individual has either inherited a weak nervous constitution as regards pain, or else that some depressing agency has lowered his power of resistance. When I speak of a weak nervous constitution as regards pain, I do not mean that it need be a generally weak physique. Perhaps a more happy word would be unstable. You remember the physicists talk of bodies being in stable equilibrium when after a disturbance they tend to return to their bottom, or centre of gravity; while unstable equilibrium is that state where a little shove off the centre, results in a big tumble. Now, the people who are markedly susceptible to pain, who have recurrences of it, may be said to have a nervous system in a state of unstable equilibrium. In other respects these same individuals may be splendid types of muscular or mental development.

The same condition holds good with pain's first cousin, muscular spasm. The analogue to the sensory crisis of attacks of neuralgia is seen in the muscular convulsions of attacks of epilepsy. And yet some of the greatest men of the world's history in mental vigor have been epileptics, notably Napoleon Buonaparte and Julius Cæsar. Although at first we may not be able to see any outward manifestation of such attacks of pain as I have spoken of, if they recur sufficiently often they are sure to leave their traces behind.

If we prosecute our inquiries in the other direction, to find what has predisposed our patient to recurrences of pain, we find in a large number of cases that his immediate progenitors have suffered from similar or allied manifestations. By allied manifestations I mean such other nervous diseases as epilepsy or chorea (St. Vitus' dance), or insanity. Moreover, there is one predisposing cause that I believe to have quite peculiar efficacy, and that is the tendency to phthisis. Again and again I have verified the truth that where a member of a tubercular family escapes consumption, he is extraordinarily liable to develop one of the graver neuroses, preferably recurrent attacks of pain.

Now, the first point we may consider settled, as to the mode in which pain expresses itself is in an inherited susceptibility, a lessened power of resistance, and this can only reside in the central nervous system.

But, as we have already said, the lessened power of resistance may be acquired, it need not be inherited.

Without stopping to dwell very long on this part of our subject, it will suffice to enumerate one or two of the principal efficient agents. And the first and far the most important of these is malnutrition of the nerve tissues, whether accompanied by the signs of anæmia and general constitutional malnutrition or not, the main cause being our civilization, with its excessive nervous wear and tear, no less in the educational period than in the intense competition of mature life. No more striking verification of this fact is needed than the results obtained in the relief of pain by physiological rest, by systematic feeding, especially of certain kinds of food, particularly fatty food. It is the general rule that in these cases there is either an indisposition to take sufficient food, or else that certain necessary ingredients are omitted owing to the patient's repugnance.

In the familiar example of sick-headache, or migraine, the patient invariably ascribes his condition to a disordered stomach, and scrupulously avoids such foods as eggs and milk and fat, which he will tell you always make him bilious. It is the hardest thing in the world to convince him that he has put the cart before the horse, and that the real fact is that the nervous trouble, the neurosis of the ophthalmic division of the fifth, is the cause and not the effect of the gastric disturbance. I am convinced that much of the suffering in the dental branches of the fifth nerve can similarly be traced to the nervous malnutrition of insufficient food, and, in addition, the local

condition of the teeth is pathologically influenced by their not getting their proper physiological stimulus in the quantity or character of the food to be chewed.

Of all the means at our command in combating the neuralgic condition, the regulating and increase in the quantity of rest and of the food supply should stand first. These facts have been known and recognized for a long time; but it is due to an American, Dr. Weir Mitchell, of Philadelphia, to have intelligently systematized their use. The principles of his treatment of nervous prostration, spinal irritation, and allied disorders, in which pain is often a prominent symptom, consists in a system of rest and forced feeding by which a larger quantity of nutriment is gotten into the system, and the waste eliminated by means of artificial exercise, by massage. It is evident that in this process the increased food absorbed into the blood goes indifferently to nourish all the tissues; but inasmuch as the muscles are not the seat of the trouble, if left alone unexercised, they would become diseased under the very stuffing process. That is where the kneading and shampooing, and movements supplied from without, are so valuable; the muscles get their healthy action without drawing on the forces of the enfeebled nervous system to set the process going. And so the nervous system has a chance to lie idle and grow fat. Similar remarkable results have been obtained in another disease whose hereditary relations to pain I have spoken of, namely consumption, by a process of forced feeding. The recent results obtained, more especially in France, by stuffing phthisical subjects, have constituted by far the greatest advance in the treatment of this disease in recent years. But in these cases the massage is entirely inapplicable because the waste of tissue is already too great. The lessons taught by the treatment of these two classes of diseases are invaluable in combating the more inveterate forms of pain.

The next acquired condition to which I would invite your attention, which may act as a cause of pain, is the presence of certain poisonous compounds in the blood or system. These are more especially the poisons of malaria, of syphilis, of gout and rheumatism, of alcohol, of certain drugs, and lastly of certain metallic poisons, as mercury, phosphorus, lead and arsenic.

Although this group includes a tolerable number of members all together, it is less important than either the preceding division of nontoxic

malnutrition of the nervous tissues, or of our first class, in which heredity plays the main role.

Still the toxic cases are sufficiently common. What we have already said as to treatment here holds good, but we must superadd the means of combating the particular poison.

In the malarial cases the pain is often entirely relieved by quinine or one of its substitutes; on this all are agreed, whether homeopaths or allopaths, or outside of any regular path. It is quite curious how the malarial neuralgias preferably locate in the first division of the fifth. But one word of caution, the mere fact of recurrence or periodicity, more or less regular, does not suffice to establish the diagnosis of malaria, for all neuralgias are apt to be more or less periodic. You must get definite symptoms of chill or fever before you can be sure. Once sure, the treatment is plain: efficient doses of the antiperiodic.

When we come to the syphilitic cases we enter more debateable territory. The pains about the head, especially the teeth, are sometimes not due to the disease, but to the means taken to combat it. I do not intend here to take up the question of the treatment of the secondary stage, except to enter my protest as to the harm done, especially to the teeth, by routine overdosing with mercury. Fortunately this abuse of a most valuable remedy is much lessening. In the third stage of syphilis you sometimes get most remarkable pain manifestations, and I had one in my practice that I cannot refrain from quoting.

A gentleman, aged between 40 and 45, had suffered for years from recurrent attacks of pain of great severity. When I saw him the pain, although more or less present constantly, had very marked exacerbations every afternoon. It was located in the great occipital nerves, especially on right side. Had formerly had considerable pain in distribution of right inferior dental nerve. His occupation was sedentary and involved considerable mental application. He stated on questioning that some eighteen years before he contracted syphilis. From this he believed himself cured. He had subsequently married, but had no children. Having suffered for six or eight years from these attacks of pain at varying intervals, he had consulted numerous physicians with only temporary benefit. He was very despondent; his sufferings were very intense, and only the most powerful anodynes gave relief. After some investigation, I made up my mind that the

syphilitic dyscrasia lay at the bottom of his suffering. I therefore began specific treatment with iodide of potash. Prof. Seguin, who saw him in consultation, concurred in both diagnosis and the line of treatment. He suggested pushing the iodide until its therapeutic limit was reached. This was done; but it was not until the enormous dose of one-half ounce thrice daily was reached that the pain yielded. During one week this patient took over one pound of iodide of potash.

A course of mercury in small doses completed the cure. Two years have now elapsed, and the patient has had no recurrence of pain.

Gout and rheumatism were formerly ascribed a much more important role in the production of pain than they now occupy. Leaving out of account the acute manifestations of these diseases, their influence is slight as predisposing causes in the production at least of facial pain. There is perhaps one disease of the dental apparatus to which I shall allude later on, in which gout may act as an efficient cause.

On the other hand, the class of pains due to the action of the chronic abuse of certain therapeutic agents is unhappily an increasing one; I allude to alcohol, opium, cocaine, chloral and other drugs, originally taken for the relief of pain, which induce a pernicious habit in their unfortunate victims, of which pain is one of the main expressions. It is an undoubted fact that this class of sufferers is on the increase. Much of this tendency is due to the excessive wear and tear and the unhealthy competition of our modern civilization. It has always been the refuge of the weak, the attempt to escape from the moral evils of our lot by means of something that will temporarily dull our consciousness of the trials we have gone through and the apparently greater trials that lie ahead of us. The moment the competition for existence and for wealth becomes keener, the greater will be the temptation of the unsuccessful or depraved to seek oblivion for their failure in some narcotic, which will for the time being quiet their disappointed consciousness. When in addition you have an inherited weakness on the part of your patient in his susceptibility to pain, or in a condition of pain actually existing, can you wonder that so many fall by the way? It seems to me that a terrible responsibility lies upon us all, especially upon us physicians, lest by our treatment we encourage this tendency. Nor do I think that as a profession we can be altogether acquitted of carelessness, to put it mildly, in this regard. It is so much easier to relieve the symptom pain,

when called to a sufferer, by a dose of morphine, and then when the next attack comes on to repeat it, than to analyze the complex group of phenomena on which that pain depends. You will perceive that the question with which we started as to the nature of pain is of vital importance in this regard.

The last group of constitutional agents which act as pain disposers is one with which you are all familiar, namely, the action of certain metallic poisons; of these the most important are mercury and phosphorus. It is highly significant that they have their main action in the structural changes they cause in the periosteum of bones, the periodontal membranes.

In the case of phosphorus, I think it is now pretty generally believed that its poison has very little effect in the mouth unless there exist a precedent caries of a tooth or its socket. These facts almost suffice to take these agents out of the group of constitutionally acting into that of peripherally irritant causes. In this class of agents, as in the preceding one, the first indication in treatment is the complete removal of the sufferer from their baneful influence.

We have now briefly reviewed the main agencies which act constitutionally in the production of pain. It is apparent, to recur to our simple illustration, that they must have their main efficiency in the action they have on the central cell, and not on any modification of the impulses sent to that cell. It cannot be denied that in rare instances these various agents are productive of pain referred to a particular nerve, when we cannot find anything in the nerve itself or in the tissues supplied by it to account for the morbid manifestation. We are, therefore, constrained to believe, at least for the present, that morbid manifestations, sensations of pain, may originate in the cell itself and thence be referred outward. But I would remind you that the whole tendency of modern medical thought is to more accurately localize the starting point of disease, and to circumscribe the area of cases in which such outward cause of disease is unknown. So long as men were satisfied to cover up their ignorance in such vague phrases as "humors of the blood," "rheumatic diathesis," etc., etc., few were tempted to carefully examine the local conditions for an explanation. But the last fifty years have seen an enormous change in our attitude of mind to these problems. It is a change which is one of the greatest in the history of the human mind. And while I do not for a moment wish to underrate the great

importance of a due regard to the constitutional causes of pain, especially of the malnutrition of the nerve cells, I believe that in the main they must be classed as predisposing causes and not as efficient ones. When we come to the question of why pain is located or referred to a particular nerve, I believe the answer in the overwhelming majority of cases will be because there is some peripheral abnormality in that nerve or in some other nerve with which it is intimately associated; for we have to recognize in the philosophy of pain the same fact that we do in the philosophy of the human mind, namely, that our ideas are so closely associated that one thought will almost necessarily suggest another. Just as, if we have always been accustomed to see Smith and Jones together, we can hardly think of Smith without Jones also putting his nose in; so in feeling sensations, certain ones get so closely intertwined that one will almost inevitably causes the other. This, then, leads us naturally to the second great division of our subject, and that is the influence of peripheral irritation in causing pain.

From what I have just said, this may be of two kinds—a reflex or associated pain expressed in some other nerve than the one affected, or else it may be due to direct irritation in the nerve itself.

A very common example of the former is seen in the headaches from which many women suffer, from the menstrual congestion (irritation of the nerves) of the ovaries and uterus. It is, however, quite outside the scope of this paper to enlarge on this curious and obscure part of our subject. I prefer to take up the more understood and more common form of direct peripheral irritation, and especially the irritation arising from diseases of the teeth and jaws.

In that delightful book, “Rest and Pain,” by Mr. John Hilton, the eminent London surgeon, he narrates a case, which is so instructive in illustrating the mode in which peripheral irritation may cause not only pain, but local disease, that I cannot forbear from quoting it:

“A gentleman, aged 63, came to consult me about an ulcer situated upon the left side of his tongue. On examination, I found an elongated, very ugly-looking ulcer, nearly as large as a bitter almond, and of much the same shape. The surrounding parts were swollen, hard, red, and much inflamed, and a lymphatic gland was enlarged below the horizontal ramus of the lower jaw on the same side. I saw in the mouth a rugged tooth, with several projecting points upon it, opposite the ulcer. This gentleman observed to

me: "Having suffered a good deal from earache on the left side for a long time, without experiencing any relief from medical treatment, it was thought that I must be gouty, and I went to a surgeon who treats gouty affections of the ear. This surgeon paid great attention to my ear, but certainly did not do it the slightest degree of good. I accidentally mentioned to him that I had for some time past something the matter with my tongue. On seeing it, he immediately began to apply caustic vigorously; moreover, not satisfied with applying it himself, he gave it to my wife that she might apply it at home. I have gone on in this way from day to day, until the pain in my ear is very considerably increased, and the ulcer on my tongue is enlarging; so I have come to you for your opinion regarding my state; for, to tell you the truth, I am afraid of a cancer in my tongue." I thought I saw the explanation of this patient's symptoms. The pain in the ear was expressed by the fifth nerve, and there was a rugged tooth with little projections on it, some of which touched a small filament of the lingual-gustatory branch of the fifth nerve in the surface of the ulcer. I detected this little filament by placing upon it the end of a blunt probe. It was situated near the centre of the ulcer, and was by far its most exquisitely painful part. This exposed nerve caused the pain in the auditory canal which led him to go to the aurist, and the aurist, instead of confining himself to his own department, seized the tongue, put nitrate of silver upon the whole of the ulcer, and increased the mischief. I simply desired that the ulcer should be left at rest; that the patient, to avoid touching the tooth, should neither talk nor move his tongue more than necessary; that he should wash his mouth with some poppy fermentation, and take a little soda and sarsaparilla twice a day. In three days about one-third of the ulcer was healed up, actually cicatrized, the enlarged gland nearly gone, and the earache much diminished.

"This rapid improvement might appear something like exaggeration, but all surgeons know that the tongue has those elements within it which contribute to the most rapid repair of injury. I do not know any tissue that repairs itself more rapidly. It is abundantly supplied with capillaries filled with arterial blood, and has an enormous distribution of nerves, and these are two elements that contribute to rapid reparation. It was quite clear that the treatment was in the right direction, viz., that of giving rest to the tongue and ulcer. After a few more days I requested him to consult a dental surgeon with respect to the propriety of taking off the points of the tooth. This was

afterwards done, and the patient soon lost his anxiety about cancer, his earache, and all his other severe symptoms.”

I cannot doubt that the starting point of a large number of similar painful ulcers and of true cancer of the mouth is to be looked for in disease of the teeth.

When we come to the teeth themselves, the pain lies in the irritated nerves of the pulp. Of course it cannot be denied that the pulp itself may be the original seat of the pain, but, if so, the number of such cases must be few. When we reflect on the mode of nutrition of the tooth, it seems almost self-evident that any depressing agency which could lead to disease of the pulp must, by an augmented action, cause greater disease in the structures which depend on the pulp for their nutrition to start with. At most, disease and pain in the pulp alone must be nothing less than a pathological curiosity. Such, however, is not the case in the vast multitude of cases dependent on caries, in which the pulp has lost in part or in whole its protection from external morbid influences. The origin of caries is one of the most interesting subjects in the whole domain of surgery. I have been astonished to find that among dentists it is not more definitely settled. So able a writer as Wm. Henry Potter (of Harvard) says: “In the first place, it may be said that caries of the teeth does not resemble caries of the bone. The term caries, as applied to the teeth, is a misnomer, given at a time when the true nature of the process was not understood.... The pathological change which occurs in caries is a decalcification and disintegration of the several tissues of the teeth.”

I confess that strikes me as a very excellent description of what surgeons usually term caries, namely, a molecular death of bone tissue. Nor can I see any difference in the essential nature of the two processes, if you make due allowance for the morphological modifications of tooth structure from bone structure, and the different environment under which the process takes place. If I were asked to define dental caries, I should say it was a molecular death of the tooth structures, especially the dentine, due to the action of micro-organisms; that in the course of the process lactic or other acid is developed, which decalcifies the teeth, is doubtless true, but the very presence of acid fermentation in a normally alkaline cavity necessitates the assumption of the action of micro-organisms. I would remind you that the

conditions favorable to the activity of such organisms are all apt to be present. They are:

1st. The presence of the micro-organism.

2d. The existence of a suitable pabulum.

3d. A certain degree of moisture.

4th. A certain degree of warmth.

5th. A certain amount of oxygen or air.

6th. A lessening of the resisting power of the tissues affected, as compared with health.

All of these first five conditions are notoriously present in the mouth. That we do not oftener suffer from their effects is due to the absence of our sixth element, the lessening of the resistance of the tissues. Thus, in health, those organisms which flourish best in an acid secretion have their baneful activity held in check by the alkaline saliva as well as the resisting power of the dense enamel. But once let the alkalinity of the saliva be lessened, or changed to an acid reaction, or let ever so small an abrasion occur in the enamel, and the myriads of these agents find a foothold for starting the morbid train of symptoms. Similarly, even in the tissues themselves, the enamel, no less than the dentine, suffers from those predisposing causes of constitutional malnutrition, which are so important in their effect on the central cell, and which act in lessening the normal resistance of the periphery.

It would be a work of supererogation to trace the progressive course of caries and the mechanism of the production of pain through irritation of the pulp. But when we come to the question of treatment, the two main considerations to be accomplished are worth our study; these are: the relief of the pain, and the arrest of the carious process. The arrest of the pain is what the patient comes to you for, and prompt action is eminently desirable. I was much impressed with this in a case I saw a few weeks ago, in which an active business man, somewhat run down by overworking, suffered from toothache (I think due to caries) for several days before consulting his dentist, my friend Dr. Jarvie. The pain in the third division of the fifth nerve gradually subsided after treatment, but was followed by a well-marked neuralgia in the great occipital nerve of the same side. He again allowed some days to elapse before sending for me, and I found him suffering from

a very intense crisis when I called. It was promptly relieved by the use of a remedy to which I invite your particular attention, namely, aconitia.

It has seemed to me for some time that this agent should form parts of the armamentarium of every dentist. From the fact that I have found it unused or unknown by some of your most progressive men, I shall not hesitate to say a few words about it to bring it before you.

Aconitia, or aconitine, is the active principle of the familiar drug aconite. Although discovered fifty years ago, it is only within the last ten or twelve years that it has been intelligently used. It is an extremely potent remedy, and must be used with great caution. In large doses it acts as a dangerous heart depressor, and paralyzer of motion and sensation. But in physiological doses it is without danger and is pre-eminently useful, because of a special action it has in relieving pain of the fifth nerve. In other neuralgias it is, for some unknown reason, far less potent. Our excellent Brooklyn pharmacist, Dr. Squibb, has put upon the market a most reliable preparation of this drug, an oleate, containing two per cent. of the crystallized salt. This seems to me a form which is peculiarly adapted to dental work. Ordinarily in prescribing this remedy internally I begin with $\frac{1}{200}$ of a grain and repeat it every hour; often one or two doses will suffice to relieve the pain, and it is seldom that more than four are required.

In using it locally if you wished to begin with the same doses, it would be necessary to dilute one drop of Squibb's oleate with three drops of bland oil for each drop of the mixture to contain $\frac{1}{200}$ of a grain.

An application of this medicament would be more effective, I believe, in relieving the pain of an exposed or inflamed pulp than those remedies now in use. I can testify from personal experience of the frequent inefficacy of the local application of oil of cloves and chloroform, while the use of the stronger remedies, as ninety-five per cent. carbolic acid or pure creosote, can only be efficacious by more or less coagulating and therefore in so far destroying the nerve tissue and the pulp. And this I take it is always to be avoided when practicable. I hope therefore, that some of you will give this remedy a trial, and verify practically my suggestion.

Having relieved the pain by one way or another, what means do you adopt to stop the progress of the caries and restore the tooth as a useful member of the economy?

Now if our considerations as to the origin of caries were true, that it is a disease due to the agency of septic micro-organisms, the logical consequence is that successful treatment must be in the line of antiseptic treatment. I presume this will cause a smile at the presumption of an outsider venturing to enter on so practical a subject, and perhaps some one will mentally quote the line about “fools” rushing in where angels fear to tread. But it is possible that much of your practice may have been truly antiseptic, just as the wise surgeon’s has been, long before we knew the why and wherefore of what experience has now taught us to be true. We are all more or less like the hero of Molière’s comedy, who was astonished to find when he arrived at middle age that he had been talking prose all his life without knowing it.

Now if we analyze your proceedings in the treatment of caries, and thus relieving the painful or inflamed pulp, let us see if they are not based on antiseptic principles, even though unconsciously employed.

First of all I take it you aim to remove all the carious material by means of your instruments, and the success of the operation is dependent on the thoroughness with which that is done. Does not that seem as if you were removing a true infective centre, and thus obviating the first condition favorable to the development of caries—the presence of micro-organisms.

Now let us see how you combat the second favoring condition, and that is the presence of a suitable pabulum; is not that done by the simple mechanical interposition of your filling between the diseased surface and the fluids in the mouth?

Again we found a certain degree of moisture needed, do you not scrupulously dry as well as clean out your cavity, and is not your filler non-absorbent as far as possible?

Fourthly. We found a certain degree of warmth was favorable; that is, of course, always present in the living body, and in choosing a good non-conductor of heat as your plugging material, it is with reference to the secondary effects of caries, the pain caused by thermal extremes, and not with special reference to the disease itself. Indeed, could we obtain a substance which would combine the resistance to organic and chemical action that gold does with the poorness of conducting power of gutta percha, it would be a great advance.

The presence of air in the decayed tooth is also prevented by the mechanical means; while lastly you substitute an artificial tissue to resist in place of the dentine and enamel that is gone.

Indeed it seems to be that the whole process of successfully filling a decayed tooth is one of the most perfect examples of antiseptic treatment I am acquainted with. I doubt not there will be further advances made in your technique, but the principles will not change. I believe it quite likely that it will prove desirable to more thoroughly disinfect the carious cavity before filling than is always done now, and it may prove possible to devise some material which, either by its hardness or by its chemical constitution, or by some antiseptic incorporated with it, will longer resist the destruction due to attrition and to chemical and micro-organic action than the ones now in use.

To recur to the main problem of our paper—the relief of pain—is it not true in this class of cases that after the first effects are stilled, its recurrence is prevented by affording artificially that immunity to the pulp from peripheral irritation which it possesses in health?

Disease of the peridental membrane causes a characteristic pain, but one which need not long detain us. From the fact that it is nearly always secondary and not primary, its treatment should first of all necessitate the removal of the originating cause. The spread of inflammation or decomposition from the pulp to the periosteum which so often occurs can be better accounted for by the hypothesis of the action of micro-organisms than by any other supposition. Moreover, in the advanced cases, where pus has formed, the same cardinal indication of treatment, viz., proper drainage, obtains here as in other departments of bone surgery.

We have already spoken of the constitutional poisons, syphilis and mercury and phosphorus, which may be causes of this form of trouble, and I would only like to say one more word, and that is in the way of treatment.

Occasionally it has seemed to me that you can stop the further progress of a periostitis, if you get it in an early stage, and prevent it from going on to suppuration. I remember one case of a gentleman who applied to me for a very painful gumboil in his lower jaw opposite the first molar. The gum was swollen and reddened, and a well-marked phlegmon could be felt. I gave him fairly large doses of mercury for a couple of days, and it gradually melted away. There was no suspicion of syphilis in this case.

Another remedy I believe to be of great value in treating neuralgia of the face starting in periosteal irritation, is phosphorus. The best form in which to administer this remedy is the preparation known as Thompson's solution. I can testify to this from personal experience. Some twelve years ago I suffered from periostitis of the first bicuspid of the upper jaw on the right side. A couple of years later, while working hard, I had an attack of intense neuralgia of the entire second division of the fifth. When it subsided, it left some periosteal thickening at the exit of the nerve from the infra-orbital foramen; and ever since then, whenever I get run down by overwork or worry, the same pain crops out. But I have found that a few doses of phosphorus will completely hold it in check; and in one or two patients, since then, I have seen the same fact, that in the neuralgia due to periosteal irritation this remedy holds a high place.

It has seemed to me highly significant that the two drugs, mercury and phosphorous, which in continued toxic doses cause this very class of diseases, should in physiological doses be curative. But this is in entire consonance with the general laws of tissue irritation, and the therapeutic fact that certain drugs acting through the nervous system stimulate in small doses and narcotize in large ones.

The last type of dental pain I will speak of is that arising from pressure due to hypertrophy of the cement. Where this is not due to the peripheral irritation of a carious tooth, the causation is both interesting and obscure. It has seemed to me that we must postulate the agency of a constitutionally acting cause to account for certain of these cases. I think it quite probable that in certain cases a well-marked gouty diathesis will be found underlying this form of disease; and a similar constitutional error must be invoked to explain the allied cases of calcification of the pulp.

We have now briefly reviewed the main forms of peripheral irritation, which act as the efficient causes of dental pain, just as we have glanced over the main constitutional causes that predispose to it. I think you will all agree with me that for the successful treatment of these cases, especially in the chronic and inveterate type, local and constitutional treatment have both to be employed. It seems to me there is great need for the more intelligent co-operation of physicians and dentists to attain the best results. Attention to one side of the question is not sufficient. For the permanent cure of our patients, the treatment of the local mischief has to be supplemented by

attention directed to the constitutional conditions that predispose to it. One of the cases I quoted illustrated the important fact that a local irritation may set up a condition of pain in other nerves which the cure of the original lesion entirely failed to relieve. This fact is borne out by many similar conditions which we meet with in other departments of medicine. On the other hand, while a particular attack of pain may be relieved by constitutional remedies, its recurrence can only be prevented by curing the local condition, which acts as the exciting cause. The enormous preponderance of cases of pain of the fifth nerve, compared with other nerves, is to be accounted for by the liability of the delicately adjusted mechanism of the organs supplied to get out of order. This is especially the case with the eye and the teeth.

It is beyond the scope of my paper to take up the various constitutional remedies of which we can avail ourselves, rest, the influence of food, the use of the various drugs, the employment of counter-irritation, of electricity, and, lastly, of those surgical procedures, exsection and stretching of painful nerves, which are our last resort.

I will more than have attained my object if I have pointed out, however imperfectly, some of the many interesting points at which our respective fields of work touch. Those points where we need your help, and you ours, to accomplish the best results.

And now, in conclusion, if we revert to our original question as to what it is that constitutes pain, I think that we will find that both the great authorities I quoted are wrong, and both are right; each has stated half of the truth.

If your observation and reasoning agree with mine, we will be forced to believe with Anstie that pain in its essential nature consists in a diminution of the vitality of our central cell, but to further allow with Erb that this is occasioned, or first brought to our notice in most cases, by an increase in the impulses sent to that cell by means of peripheral irritation.

BACTERIA, WITH A METHOD OF STAINING FOR DIAGNOSTIC PURPOSES.

BY JOSEPH KETCHUM, ESQ.

Read and Demonstrated before the Section on Microscopy of the Brooklyn
Institute.

In presenting the subject of Bacteria, I wish to disclaim any originality for the matter offered. I have endeavored to collect from such sources of information as I have access to the important dates, names and facts which have marked the progress of bacteriology up to the present time.

So far as we know, the first observer of bacteria and the so-called infusoria was Leeuwenhoek, who, with a simple magnifying glass, noticed in a drop of putrid water the multitude of little granules moving about in it. This was in 1675, and his observations were communicated to the Royal Society of Sciences in the same year. In the following year he recognized bacteria in the tartar from the teeth, and though he did not name them, his description of their forms and his drawings enable us to identify them as vibrios. There appears to have been no important investigations carried on until nearly one hundred years later, or in 1773, when Müller, a Dane, attempted to classify the organisms then known. He called them all infusoria, from the fact that they were the product of infusions, and divided them into two genera—the monas and vibrio. The monas he subdivided into ten forms and the vibrio into thirty-five; but his descriptions of them are so faulty that it is at present impossible to identify them from his writings. During the following century the study of bacteriology attracted more or less attention, and in 1829 Eherenberg, who is the Humboldt of the science, commenced his investigations, which for fifty years he pursued with an ardor and enthusiasm second to not even Darwin himself. He, in 1838, classified the family of vibrioniens, and with the additions made by Dujardin in 1841, placed them in a scientific category. Of course during this period many were the disputes and discussions as to specie, genera or family, each newly discovered member belonged to. And we have to come to the period of Hallier, Hoffmann and Cohn, and many others, before the

questions, which had up to that time been in dispute, were settled. Ehrenberg's original classification was into:

1. Bacterium, or rod-like—three species.
2. Vibrio, snake-like and flexible—nine species.
3. Spirillum, or spiral, but inflexible—three species.
4. Spirochæte, spiral, but flexible—one species.

Dujardin, in 1841, in his *Natural History of the Zoophytes*, accepted the classification of Ehrenberg, except that he unites the spirillum and spirochæte, calling them all spirillum. Up to this time all bacteria had been considered animals, but a close study of their life history and habitat by those who followed declared them to belong to the vegetable kingdom, and as such they are accepted to-day.

In 1853, M. Chas. Robin pointed out the relationship of bacteria to Leptothrix, a form of fungi closely allied to that of mildew; and M. Davaine, in 1868, clearly demonstrated their relationship to the vegetable world. From this time the progress of bacteriological investigation has made rapid strides. Prof. Pasteur in the organisms of fermentation and the role they play therein; Davaine and Hallier in demonstrating the specific relationship of bacteria with charbon or anthrax; and the work of Koch, Nageli, Kohn, Billoth, Miguel, Burdon, Sanderson, Klein, Weigert, Klebs, Ehrlich, Sternberg, and many others, are too recent to require special mention.

Few have more than the faintest conception of the minuteness of these organisms. Prof. Cohn, justifying himself for the unscientific method of comparison which he uses in class instruction by Prof. Tyndall's argument on the scientific use of the imagination, says he compares man to the cheese mite, as the Strasburg cathedral to a sparrow. Of the animalcules which Leeuwenhoek discovered, they are to man as the bee is to the horse. As improvements have been made in microscopes, just so fast have we penetrated into the world of micro-organisms, until now the proportion between the smallest we can see and man, is as man is to Mont Blanc.

Of course, with these exceedingly minute structures, nothing can be made out except points. Among some of the larger forms, a few have been able to see cellia, and in some cases the growth of the spores; but in the

present state of microscopical optics the work is slow, and progress in this direction is waiting an advance in the science of optics.

Like all living organisms, bacteria propagate themselves. The most usual method is by fission or by partition, though Magnin and Cohn have recorded their observations on the formation of spores and sporangia, and I have myself witnessed the last named method. It is of importance to note that while the bacterium is killed by continued exposure to temperatures of freezing or 176° F., the spores will germinate after protracted exposure to temperature as high as 205° F. or as low as 123° F. These spores will also withstand complete desiccation, and it is in this form, mixed with the air we breathe and move in, that present the conditions from which all zymotic diseases originate. Miguel has shown that, while the air contains very few adult bacteria, it contains myriads of their spores. To the researches of Koch, Pasteur, and others, we are indebted for the certain information that, while these omnipresent germs withstand such vicissitudes of temperature, they require certain food for their maintenance; and though we cannot as yet tell what that food is, we know that when nutrient material is submitted to their action they thrive for a time, and when the particular principle which supports them is exhausted they die. This is particularly true of pathogenic germs, and the accepted theory of the bacillus tuberculosis, or the germ of consumption, is a good illustration. It has been demonstrated by Koch, Klein, Pasteur, Frankell, Sternberg, and others, that they require some product of inflammatory action for their support within the body of their victim. This is also true of cholera, at least so far as their dietary requirements are concerned. The animal cannot be infected with tuberculosis by merely introducing the germ-laden material into the stomach or upon any of the mucous membranes; but if an inflammatory condition be present, either due to the puncture of the introducing needle or scalpel, or to extraneous causes, such as a catarrhal condition of the lungs, tuberculosis is as sure to follow as the sun is to rise again.

The human mind can scarcely comprehend the enormous numbers of these omnipresent atoms without a resort again to the legitimate use of the imagination. A computation of the increase from a parent germ shows as follows: We know that the parent grows until it reaches double its original size, when it constricts itself in the middle like a figure eight and breaks into two individuals. Each of these divides again, and, on account of the rapidity with which this is done, we find them usually in chains or squares.

The warmer the air, the faster this proceeds, and at the temperature of the body the entire life history of a germ, from the time of fission of the parent to the time of his own subdivision into two new individuals occupies less than one hour. This gives us a known quantity for our problem. Let us look at the result. From a single germ increasing by the power of two each hour, we have at the end of twenty-four hours 16,777,220; at the end of two days the number has increased to 281 billions, and in three days to the enormous number of 48 trillions, and in one week the number can only be expressed by figures of fifty-places. In order to make this number comprehensible, let us figure the mass and weight of this, the result of a single bacterium. A single Bacterium Termo has an average width of $\frac{1}{1,000}$ mm. A cubic mm. would therefore contain six hundred and thirty-three millions, and in one day would be one-fortieth full. At the end of the following day there would be required 444,570 such cubes to contain the product of the parent, or say half a litre. Suppose the seas of the earth cover two-thirds of its surface with a mean depth of one mile, the aqueous product would be 929 million miles. Now, our parent germ and its product would in five days completely fill this space. More wonderful still is a gravimetric estimation. Suppose we call the specific weight of the parent germ the same as water, which cannot be far from right, it would appear that the parent weighs, or his equal bulk of water weighs, 136 millionths of a gramme; in forty-eight hours, 442 grammes; in three days, nearly $7\frac{1}{2}$ million kilograms; and, inside of thirty days, the weight of the earth itself.

Prof. Cohn, in offering these figures, says: "I don't consider this idle play; without it we can form no conception of not only the enormous increase, but the tremendous destruction of these germs which is going on around us. Food is lacking to support more than a comparatively small proportion of the product of the parent, and, as it is demonstrated that they feed from their environment, one can readily understand that without a constant supply a given infectious germ will with its followers soon destroy its nidus or perish from starvation."

Our breweries demonstrate the truth of this hypothesis; for, in twenty-four hours, a single yeast cell, which is $\frac{8}{1,000}$ mm. in diameter, will yield one hundred-weight of yeast.

I have endeavored to present the subject in a condensed but general way without burdening you with technical details of species, genera or life history. The subject is a vast one and to which the best minds of the scientific world are devoting themselves. To those who are or may become interested in bacteriology and particularly to those who study the relation of these germs to disease, is held forth the reward which is sure to come to those who work persistently and intelligently.

The method which I shall employ to-night is eclectic. Doubtless each investigator will find fault with some parts of the process and perhaps suggest a better one. The following, however, has in my hands worked well and given entire satisfaction, so far as I know, to those who were and are most interested.

The apparatus necessary is as follows:

One two-inch glass funnel.

One package filter papers to fit same.

Four medium size test tubes.

Two glass or porcelain staining glasses.

One glass or agate mortar and pestle.

One cover holder.

One pair pincetts.

One alcohol lamp.

Package of wooden toothpicks.

The cover holder may be easily made by taking a piece of thin platinum, two inches long and one-eighth wide, splitting one end for half an inch up and bending into a Y shape, then lashing to a small handle (I use a match). This little tool is most convenient for floating cover glasses in staining fluids.

The reagents necessary are as follows:

A five per cent. solution of nitric acid in alcohol (95 per cent.).

Saturated alcoholic solution of fuchsine.

Saturated alcoholic solution of methyl blue.

Small quantity of alcohol, 80 to 95 per cent.

Pure colorless aniline oil (anilin).

The method is as follows:

First pour enough aniline into a test tube to cover the bottom and half fill with water, shake violently for two minutes, and filter through funnel, which has previously had wet filter paper fitted. It is essential that the filter paper be saturated with water, else the aniline oil will separate during filtration. Our next step is to deposit specimen of sputum in mortar (if very viscid, add a few drops of water), and triturate thoroughly in order to break up encapsulated colonies, and distribute evenly through the specimen.

Now remove an amount which will just cover end of toothpick, and deposit it on a previously cleaned cover glass, which should not be over $\frac{1}{100}$ inch thick, and thinner if possible; immediately cover with another cover glass, allowing sputum to spread by capillarity or slight pressure, and separate by sliding apart, and put aside to dry without heat. I have found that specimens dried without heat (and consequent coagulation of albumen) will show a much larger number of bacilli than when heat is used. I believe this is due to the fact that the fuchsine penetrates more thoroughly through the albumen when not coagulated, or that when it is coagulated by heat it to a greater or less extent it protects them from the action of the stain. While the covers are drying we will pour out a sufficient quantity of the aniline water, which by this time has filtered into one of the staining glasses, and add one or two drops (not more) fuchsine solution. Now, placing one of the cover glasses on our cover holder, sputum side down, we lower it into the staining fluid and withdraw holder from the side, and repeat the operation for the other cover glass. It is my habit to allow the covers to remain in this solution for at least eight hours or over night. The time may be reduced to ten or fifteen minutes by heating the red stain to about 140 or 150 F., but the result is not so brilliant, nor is it sure, as I have frequently failed to find the bacilli by the short method, but have been able to demonstrate their presence by the long one.

At the end of either of the above periods of time, the cover glass is lifted out of the staining solution and, without washing, immersed in our five per cent. solution of nitric acid and alcohol. It is this part of the process, if any, which will give trouble, as the time of immersion is governed by the thickness and general character of the sputum. My custom is to hold the first cover immersed until the color has just disappeared, or say fifteen

seconds, and the second five seconds longer; but a very little experience will remove any difficulty from over-decolorizing.

From the decolorizing solution they are immediately immersed in water and thoroughly washed, when they may be again floated in the contra-stain, which is prepared by filling the other staining glass with water to which a few drops (three or four) of our methyl blue has been added. They should remain here for from five to eight minutes, when they are again removed with the pincetts, and a few drops of alcohol poured over them to wash off the surplus stain. Again wash in clean water, and dry by gentle heat (which will now do no harm) over the alcohol lamp, and place sputum side up on table.

A very small drop of thin benzole balsam is now placed in the centre of each cover, and a cleansed slide gently lowered over one in such a position that both covers may be mounted on a single slide. As soon as the slide has been sufficiently lowered to come in contact with the drop of balsam, it spreads by capillarity, and draws the cover close to the slide without the slightest danger from air bubbles being engaged, and the slide may at once be inspected by a *dry* objective.

I have found it necessary to use an objective at least as high as one-fifth or one-sixth, with central illumination without diaphragm, as cases will frequently occur where the staining is so faint, that with a lower power they will escape observation, though a good, wide angle, four-tenths inch, will show them well when strongly stained.

I have endeavored to explain the method with perhaps too strict a regard to detail, but am sure that one who follows the various steps once or twice cannot fail to acquire the necessary technique without occupying more than fifteen minutes of working time; that is to say, five minutes to the first staining, and then the following morning to prepare and mount for observation.

171 GATES AVE., BROOKLYN.

**ADDRESS TO THE GRADUATES OF THE LONG ISLAND
COLLEGE HOSPITAL TRAINING SCHOOL FOR NURSES,
DELIVERED JUNE 12, 1888.**

BY GEORGE G. HOPKINS, A.M., M.D.

Ladies and Gentlemen and Class of 1888: We are apt to claim the trained female nurse as the outcome of the more rational treatment of disease, in modern times, but this is wide of the truth. So far as I can ascertain, in my researches among the ancient Vidas of Hindostan, and the literature of Egypt, Greece and Rome, I find no allusion to female nurses as a class, until the third century of the Christian era. Surgery and medicine had attained a high degree of perfection, many operations which to-day we claim as new to the nineteenth century were successfully performed 4,000 years ago; but the special nursing of them seems to have been done by the medical student, or by the practitioner himself. The earliest record I can find of women devoting themselves to the care of the sick, and attending to all the duties of a trained nurse, is that of Empress Helena, mother of Constantine the Great. This noble woman, who lived nearly fifteen hundred years ago, not only founded a hospital and endowed it, but herself, with the ladies of her court, there gave the most devoted and tender care to the sick poor. The Emperor Valens presented the most beautiful grounds and buildings in the neighborhood of Cæsarea to Archbishop Basil, “for the benefit of the poor whose bodies were afflicted with disease,” as being those who stood most in need of assistance. And as early as A. D. 373, the Archbishop had organized at Cæsarea an immense hospital, called the “Basilides,” which Gregory Nazienza thought worthy to be recorded among the wonders of the world; so numerous were the poor and sick who came thither, and so admirable was the care and order in which they were served. The charge of these sufferers was not at first assigned to humble hands; the most illustrious ladies of the empire participating in the offices of mercy.

At Constantinople the Empress Flacilla, wife of the elder Theodosius, in the year 380 was watching with solicitude over all those whose bodies were mutilated, or who had lost limbs. She visited them in their own dwellings, waited upon them herself, and supplied their wants. She repaired with the

same zeal to the public hospitals of the church, where she attended the sick, made ready their culinary utensils, tasted their broth, carried the dish to them, broke the bread, divided the meal, washed the cups, and performed for them all the offices which usually devolve upon servants. One might justly be proud to be in such royal company, and regard, as she did, nothing degrading which is necessary to be done for a sick patient.

In modern times, the revival of nursing by trained women is due in great measure to that noble and accomplished woman, Florence Nightingale. As early as 1844, at the age of twenty-one years, she began to exhibit her interest in and the alleviation of suffering, and the improvement of the care of the sick poor in the hospitals of Great Britain. She visited and inspected the hospitals of Europe, and in 1851 entered into training as a nurse, in the institution of Protestant Deaconesses, at Kaiserswerth on the Rhine. On her return to London she put into thorough order the Sanitarium for Governesses, in connection with the London Institute. She served ten years of apprenticeship before entering on her life work.

In the spring of 1854 war was declared with Russia, and an army of 25,000 men was despatched to the Crimea. The faulty arrangements of the British government for the care of the sick and wounded furnished the theatre in which Florence Nightingale was to win her first laurels. The hospitals were soon crowded, and the mortality in the wards so great that the casualties of the fiercest battles were as nothing in comparison.

The war office recognizing the condition of affairs, gladly accepted the offer of Miss Nightingale to go to the seat of war and organize a nursing department.

Her devotion to the sufferers can never be forgotten, she has stood twenty hours at a time, directing and assisting in the care of the sick and wounded. Her unfaltering devotion and incessant work undermined her health; but though sick and feeble, she never left the field of duty until Turkey was evacuated by the English troops. Major Delafield (who with Maj. Mordecai and Capt. Geo. B. McClellan, U. S. A., had been sent to Europe by our government, to study the art of war in the Crimea), in his report to the War Department, remarks, in speaking of the English hospital at Scutari, "It was in this well-arranged hospital that that most estimable lady, Miss Nightingale, exercised her powerful influence in alleviating the condition of the sick and wounded from the battle-field. Women as nurses were

employed to attend upon the men in the wards, under the kind and beneficent guardianship of this good lady, with the many advantages that would naturally follow the most gentle, painstaking, and cleanly attendance of women as nurses. Miss Nightingale's efforts have resulted in the establishing, in connection with the English army, an office known as the 'Superintendent General of Army Nurses,' the office to be always filled by a woman. She has under her a corps of female nurses, who take care of the sick in the military hospitals." The Sanitary and Christian Commission of our late war was the outcome of the volunteer nursing in the English war of the Crimea and the fruit of these efforts in this country are the training schools for nurses which have sprung up all over this land.

Next to our entrance into this world and our departure from it, occasions such as the present, when we have completed our education and are about to enter upon our chosen vocation, are the most important events in our lives. The calling which you have chosen, while not a new one, is comparatively new in having special schools, and courses of study provided for it. Nursing has always been considered peculiarly woman's work—more or less adaptation to such work is inborn in woman. What man can smoothe the pillow of the sick, or soothe an aching brow as gently and acceptably as one of the gentler sex! Who can move as quietly, and approach the bed of pain so gently as woman!

I have seen sick men, absent from home and friends, sigh for a mother, sister, or wife who is not at hand.

Thanks to this school, and others, everyone can now have skilled female care when sickness and disease are upon them.

You who are about to go out from us to-day, are entering upon a calling which will require all the skill, faithfulness, courage, patience, forbearance, endurance, watchfulness, self-possession, tenderness, cheerfulness and tact, that a human being can possess, and above all, "a conscience void of offence toward God and man." "To thine own self be true, and it doth follow as the night the day, thou canst not then be false to any other." You have each and all of you received, at the hands of your admirable Superintendent, and the lecturers of the College, such definite and varied information in all the departments in which you may be called to act, that you ought to be prepared for most emergencies, and have shown by your examinations that you have heard and understood them.

The fault will be yours, and yours alone, if you do not treasure them in your minds, so that you may be not only trained but *skilled* nurses. The responsibility for the proper management of a sick-room and the patient in it is a very high and grave position, and requires the utmost faithfulness on the part of the nurse. Unless you are willing to put aside everything that may interfere with your giving *yourself* entirely and conscientiously to the care of your patient, do not assume the charge. But when you once receive that charge remember that you are dealing with that which disease can destroy by your negligence, and no human power can restore—a precious human life. You therefore owe to each case all that a faithful mind can suggest and the body endure; and such faithfulness has not been wanting in the former graduates of this school.

I can never forget the scene when one of our graduates, after having charge of over thirty cases of typhoid fever among some orphan children, and we had to lose one, whom she had nursed as faithfully and tenderly as its own mother could have done, how, when she had done her all and death claimed him, there were tears shed for one who had no mother to shed them. And it was due largely to her unremitting faithfulness that we were able to record but two deaths in thirty-three cases. That woman has not had an idle day to my knowledge in several years.

When any unexpected emergency arises, which to your educated eye teaches you that your patient is in extreme danger, do not alarm the friends unnecessarily; try not to show in your voice or conduct that you are demoralized and have lost courage. While doing the best you can (until the arrival of the physician, whom it was your first duty to have summoned), encourage those around you, and keep them busy if you can, as, unless they are occupied, the coming of the physician will seem to them unreasonably delayed, even though he be at hand when called. Above all, do not let every physician within reach be sent for, unless the situation is one of great urgency, as I believe many patients have been frightened to death by the demoralization of solicitous friends.

If you can show yourself capable and maintain your own composure of mind, you will be able in nearly every instance to avert a panic, and in many cases prevent disastrous results to your patient. But if you fail at times for want of proper courage, do not give up with a feeling that you are

unequal to emergencies, only be the more determined not to show the white feather again.

It is a well known fact to military men, that veteran troops who have stood the brunt of the fight in many a battle, become demoralized unexpectedly and retreat, to the utter surprise of their officers. But in their next battle their courage and deeds of prowess again surprise every one. So be it with you.

In the sick-room nothing so distresses the helpless sufferer as a want of frankness on the part of the attendant. You may refuse or neglect to answer, or turn the subject if possible, but never tell what are called “white lies.” One lie always requires another to cover it, and sooner or later you will be caught. If it is not best to tell, say outright it is better for me not to answer that question; or it may be the least of two evils to answer it faithfully as patients often imagine that things are far worse than they are. I believe that we of the medical profession often err in withholding from patients that which it is best in the end that they should know. This is one of our most difficult lines to draw.

If you have not learned or are not determined to learn to endure the caprices and demands of unreasonable men and women in the most unreasonable hour of their lives, you have mistaken your calling; as without Christian patience, I do not believe it possible for a nurse to succeed for any length of time. The trials and vexations of a nurse’s life are so numerous and so constant, that it is a wonder to me that there are so many who are ready to enter this calling in life. There is no need since the revival of professional nursing for women to torture themselves or do penance. Be as ready to minister to disagreeable people as a person who two years ago wrote that she would take a fresh air child, saying “send me one of the dirtiest, most unattractive and unruly of the children, one whom nobody else wants.” An unreasonable, selfish and wilful patient is a purgatorial discipline for both nurse and doctor.

Remember that the most gentle and considerate of people will say and do things when sick unwittingly, that in health they would sooner cut out their tongues or destroy a limb than say or do. The mind is sick as well as the body, and the patient not responsible. Cultivate forbearance and endeavor to sear all your tender points. Be ever ready to excuse and believe that no slight was intended, unless it is reiterated and you are forced to believe it.

The physical and mental strain which you are at times called to bear will be very great; that you may be able to endure it, you must give special care to your health. You have been taught the laws of health, and yours is so arduous a calling you must observe them strictly. Dame Nature is a stern mistress, and if you disobey her you will surely suffer for it. When you are out of employment you will need recreation and diversion to keep both body and mind in the best condition. When you are in charge of a patient, the time away from the bedside is not yours to do with as you please, but for rest and fresh air; as you owe it to your patient to give the best possible service, and thus only can you do it.

You must cultivate the habit of observing the least change in your patient's condition, so as to be ready to meet any emergency; it will not do to sit down and watch your patient as a cat would a mouse. Yet in severe cases your eye should hardly ever be off your patient; this should be accomplished and can be done in such a way as to be almost imperceptible to the sufferer. Every little change should be noted, and if any importance may attach to it, it should be written down as soon as you can conveniently do so. You are the physician's eyes, ears, and hands while he is absent; you cannot therefore be too watchful.

Each one of us has certain vulnerable points of character, but it is not always easy for us to see them. If we would be self-possessed we must seek to discover these weak points in our armor by seeing ourselves as others see us; then by learning how to cover them, and not be disconcerted when our weak point is attacked. No *one* virtue is of more value in your arduous calling than this one of self-possession.

In this world of care and trouble much can be done to ameliorate suffering and soften the sting of pain by tender, sympathetic care; your patients will expect less of you if all you do is done with ease and quietness and thoughtful tenderness. You will then be likely to gain a friend in every patient; the patient will feel that a friend has gone when you depart.

A cheerful character rides smoothly over many rough places in this world that otherwise would jolt terribly. A bright, cheery nurse is better than many a dose of medicine for the patient; therefore be always cheerful. By cheerful I do not mean frivolous, as levity is the last thing that should appear in a room where such mighty elements are at work as in the sick-chamber. Therefore be cheery, but not mirthful or giddy.

There are some words in the dead languages which it is almost impossible to put into English without, in a great measure, losing their meaning because they contain so much in themselves; they are so difficult to define. So there is one little word in the English language that contains so much in itself that it is impossible to define it in a few words, and after using many you feel that you have only sailed around it without getting at the central and most important part of it—that word is *tact*. But it is the want of that which has consigned some of the brightest and noblest minds that I have known to oblivion. I call to mind just now one of the best read and most highly cultured and gifted men that the medical profession of Brooklyn has ever known. He lived and died among us, unappreciated except by the few who knew him best, little sought after by those who needed balm for their diseases, which he was better able to apply than most of his companions, and with scanty maintenance, while medical sky-rockets about him were riding into lucrative practices. The suffering continued to suffer, when, if they had only known it, skilled and efficient help was at hand, in a man who did not know how to so bear himself as to win the confidence of the community. Had he possessed a little tact his name would have been known to the world.

I want to say to each one of you, consider well if you propose to follow this arduous calling, pause and consider whether you really feel that it is your vocation, and feel equal to its physical and mental demands.

An ideal to strive after is good for us all. I will lay before you to-night one that was realized in the history of a friend who is now in a better and happier clime than this, and whom I would be glad to have each one of you strive to emulate.

Some years ago, before, as far as I know, there were any trained nurses in this city, I was asked to go to see a lady in a neighboring village, who had been confined to her bed for more than a year, and was supposed to be incurable. A year from that time she was able to be about, and six months later she determined to devote her time to the care of the sick poor. She did so, and I never had any one who would, or could, take better care of every case that fell to her charge. I always felt that, as far as human skill and strength could do it, my directions would be carried out to the very letter. Her last case was that of a little girl who had been burned over about three-quarters of her body, a degree of burning usually considered fatal; but in

this case it did not prove so; and for months this noble woman dressed this suffering child, and would let no one else do it. Little Tina dreaded to have any one else touch her. The child was almost well, and this good woman was just finishing her morning dressing of the burn, when she suddenly fell back and expired. The soldier died at her post of duty.

“Like a star which maketh not haste and taketh not rest, let each be fulfilling his heaven born hest.”

THE ETHICS OF OPIUM HABITUÉS.

BY J. B. MATTISON, M. D.,

Read before the Society of Medical Jurisprudence and State Medicine, June 14, 1888.

“All men are liars,” said the writer of ancient days, and the revised version of modern times is, “All men—who take opium—are liars.”

The writer—whose initial acquaintance with this question dates back nearly two decades, and whose professional experience for several years has been exclusively devoted to a large and enlarging clientele of this class—has long held this opinion to be a mistaken one. Years ago he wrote—“Clinical Notes on Opium Addiction,” read before the Kings Co. Med. Soc., 16th January, 1883—“Nor do we share in the opinion, largely held, that no reliance is to be placed on the word of opium habitués. That the habitual use of opium, in many cases, does exert a baneful influence on the moral nature we are well aware, but we also know that in the ranks of these unfortunates are those who would scorn to deceive, and whose statements are as worthy of credence as those upon whom has not fallen this blight.” Increasing attention to this topic has only confirmed that belief, and the recent statement—unwarranted and untrue—of a medical writer and teacher, that “no morphia habitué can be depended on to tell the truth,” with the courteous invitation of your honored President to present you a paper, has prompted me to offer some thoughts on this subject—the result of observation, reflection and applied common sense.

Putting the query—why do men take opium?—the answer to-day is that made nearly twenty years ago by Dr. Joseph Parrish, Pres. Amer. Assoc. for the cure of Inebriates—“men take it for a physical necessity.” In an experience covering the history and treatment of hundreds of cases, I have noted only two exceptions.

Let it be distinctly understood that my remarks apply only to the better class of habitués, who have become such by force of conditions beyond control. With those who, viciously indulgent and lacking alike in principle and purpose, take opium from mere sensual desire, we have nothing to do.

This physical necessity, the great genetic factor in an opiate using, it need scarcely be said, has its rise in painful disorder of body or mind. For this opium in some form is given, which, when the legitimate need for its action is ended, entails a demand for continued taking that will not be denied.

The larger share of responsibility then rests on the medical man who prescribes—very properly it may be—this valued drug, though the main measure of his responsibility depends not on the initial using, but upon the case being dismissed without full thought as to the ultimate result of the opiate taking, and with a neglect to warn the patient against the danger of continued using, and insisting upon—giving to this his personal attention—the entire narcotic disusing when the proper need for its taking is ended. Vide “The Genesis of Opium Addiction,” *Detroit Lancet*, 1884, and “The Responsibility of the Profession in the Production of Opium Inebriety,” *Med. and Surg. Reporter*, 1878.

Granting this correct, on what principle of equity or right can one be held accountable, and so culpable, for his use of the drug when, unaware of its ensnaring power, and, confiding in the counsel of his medical adviser he avails himself of the relief it affords?

Another and most important auxiliary factor obtains in these cases—one of which the laity knows little or nothing, and the profession appreciates less than it should—and that is the power opium possesses to create a necessity of its own. Of this, I venture to assert that no one, other than the subject of a painful personal experience, or of large observation, can form a fully adequate idea. The writer has been studying opium and opium habitués for more than sixteen years, with an annual experience, of late, as regards number of cases, that is probably unequalled in this country, and yet he stands more and more in awe of this peculiar power with every case that comes under his care.

Granting a painful physical necessity, and the daily or semi-daily use of opium—especially morphia, subcutaneously—for a few weeks or months, and there are few, if any, who can withstand the ensnaring, enslaving power of this drug. Men stronger of brain and brawn than we have gone down before it. I have known a superbly athletic specimen of physical manhood, able to resist the wintry rigor of a polar expedition, succumb to the power of morphia in less than a month. I have seen a man so generously endowed that he survived the horrors of Salisbury when the death rate averaged

eighty per cent., go down before the same resistless power in four weeks. It was my pleasure to see this gentleman recover, and take the lecture platform to tell of his bondage and escape, and this is what he said:

“I proclaim it as my sincere belief that any one afflicted with neurotic disease of marked severity, and who has in his possession a hypodermic syringe and morphia solution, is bound to become, sooner or later, if he tampers at all with the potent and fascinating alleviative, an opium habitu  . The first dose is taken, and mark the transformation. This overmastering palliative creates such a confident, serene, and devil-may-care assurance, that one does not for once think of the final result. The sweetness of such harmony can never give way to monotony. Volition is suspended. You may not think of it when the pain for which it was taken subsides. But when distress supervenes you go at once for the only balm that abounds in Gilead, and every additional dose is but another thread, however invisible, of which the web is made that binds us fast as fate.”

If this be true—and it is true—what justice is there in the charge that these unfortunates continue the use of opium from an innate propensity to evil, or a merely vicious desire? What right have we to set ourselves up in judgment to note the beam in our brother’s eye, when the only reason it is not in our own—when the only reason you and I are not opium habitu  s—is because a kind Creator has so conditioned us that this physical necessity, and consequent opiate need, does not, with us, obtain?

Having thus touched upon the etiology of this disorder, let us reason together regarding the special ethical point involved, and note the reason—if reason there be—for the commonly accepted idea that all men who use opium are liars. While admitting that the habitual need and use of this drug does, in many cases, warrant such assertion, I hold that the leading factor in this moral obliquity is the principle of self protection—the habitu  ’s desire to shield himself from that censure which the prevalent opinion—uncharitable and untrue—that he is simply the victim of his own vicious indulgence, involves.

There are various proofs that this holding is correct, and, too, without resorting to the opinion held by Lahr, Fiedler, and some others, that opium habitu  s are the subjects of a mental alienation, both in the creation and continuance of their addiction, and therefore absolved from culpable wrong—an opinion in which I do not share. Nor do I believe, for reasons given,

with another German observer, that “the morbid craving for morphia ranks among the category of other human passions, such as smoking, gambling, greediness for profit, etc.,” for if this were true, the impulse to protect one’s self would not so largely prevail.

The opium habitué realizing that he is looked upon as one who has given himself up to a vicious habit, a habit in which he persists from mere desire to enjoy the pleasures of opium—pleasures which, be it ever remembered, soon give place to its pains—and so liable to the censure which a vicious indulging involves, is impelled—by a feeling common to us all of guarding our good repute—to yield to the protective temptation to untruth.

But to this there are numerous exceptions, for many a captive to this drug, though well nigh crushed by his captivity, and that “cruelty of ignorance,” which the unjust reproaches of should-be friends entails, still refuses to seek refuge behind such subterfuge, and scorns to tell a lie.

And do we not note this same impulse to deceit in most non-habitués who, lapsing from the right, make effort to avoid the sequence of their sin? Does the swindler always confess his swindling? Do the thieves, the forgers, the rascals of any degree, never deny their wrong doing? And while, in these cases, such double wrong may be the outcome of a general depravity, that, of itself, tends to prove that if the pernicious effect of opium in this regard were due solely to its baneful effect upon the morale in general, we should note the same tendency to lying along the various lines of life, whereas, it is a fact that on any question other than one involving his opiate taking, and consequent accountability, the habitué may be, and often is, a very prototype of truth.

What is the bearing of this question on the medico-legal status of these cases? If they be held culpable for the inception and furtherance of their condition, whatever outcome there may be affecting the jurisprudence of their action, must, from such erroneous view of the situation, fail of that legal justice which a correct appreciation of their case demands.

The writer was recently called upon to testify in the case of a physician who had been under his care for treatment of narcotic inebriety. This gentleman was the subject of delusions and hallucinations, so marked, that, in my opinion, he was not accountable for his conduct. Suit for separation was brought against him, and the referee’s remarks during the trial, and his

final decision, were in keeping with the belief that the defendant was responsible for the consequences of his alienation.

Again, a right appreciation of the status of such patients will lessen the labors of the doctor and the lawyer in their legal aspect, and remove the risk of failure to determine the true physical condition of the habitué where the question of narcotic taking is the leading issue in the case. To illustrate. Granting a general acceptance of my assertion that the class to which this paper pertains are the subjects of a disturbed organism, beyond their control, and for which they are blameless; and granting an appreciation of this belief by the patients themselves, then their main motive for concealment will be removed, and no more reason for untruth exist than if they were the victims of any other functional disease.

Again, the present general opinion of these patients is such that once a case comes into court to settle the question of an opiate using, the defendant, desirous of protecting himself, by denying his drug taking, makes it essential that evidence be secured to disprove his statement, and if certain signs be wanting, the habitué may quite outwit the medical expert. The writer noted a case of this sort last summer. A lady, cultured and refined, who had fallen a victim to morphia years before, and who was party to a suit in court, was examined by two well-known female physicians of this city, who, failing to apply the one infallible test of an opiate using, testified that she was not an habitué. They were mistaken—the lady was taking morphia, though she has since recovered. The point involved, to spare the chagrin of such an error, is obvious.

Lastly, what is the trend of a more rational view of this question as regards the treatment of these cases? Reference has been made to the statement that “no morphia habitué can be depended on to tell the truth”—a statement so often at variance with the fact that it must be the outcome of an experience with the baser class of cases—and I submit the wrong of regarding *all* as liars because *some* fail to tell the truth; or, added evidence of the “cruelty of ignorance;” or, an unwarranted libel on a worthy class of unfortunates, who, Heaven knows, have enough to bear without loading them with the reproach such an injustice implies.

Nevertheless it is just such an opinion, and consequent lack of confidence in the honor of these patients that influences their management by some medical men. Looked upon as the victims of their own wrong-doing, or as

unworthy the sympathy that should ever exist between physician and patient, or treated on the erroneous belief that such is the only proper method, they are consigned to the brutal ordeal of abrupt and entire opiate disusing, which, while it may end in the desired result, entails such suffering of mind and body as to be utterly inexcusable—because a more humane method will avail—except under conditions peculiar and beyond control.

I am well aware that such coercive measures are the only hope of cure in some cases, but I also know that such patients are not of the better class, and that, once the drug abandoned, the prospect of continued recovery is small, because they lack one of the essential requisites for a permanently good result—that is an earnest desire to be cured.

And the promise of good results from this better way in regarding such patients is more far reaching than on first thought might appear, for the ex-opium habitué forced to stem a tide of distrust—special, as to his cure, and general, as to the permanence of that cure—finds himself hampered in continued well-doing by the lack of that hopeful trust that would largely conduce to his good getting on.

In a recent letter from a lady who honored the writer with her care, nearly three years ago, she referred to the permanence of her recovery, and added —“but as I found it difficult to make every one believe this, much less acknowledge it in my favor, I resorted to the best means I could think of to establish corroborative testimony that *would* avail, and during *all* the time I have been in or near—the past two years, I have gone regularly every few days to a physician of prominence here, my old friend and medical adviser of many years’ standing, and had him make every test he desired, placing on record my exact condition, and showing the real truth of the matter. I continue to do this, and intend to do so, and have let people generally know that such a record is being made. I need not tell you that I am proud of my victory. The struggle against ungentle and unfair judgment of those around you make a combination of overwhelming power against the reformed opium taker. It is there that the *real* conflict begins.”

The writer’s professional work among this class has long been along the line here noted. He has extended confidence—very rarely has it been broken; he has asked for confidence, and the general result can be truly and tersely stated—increasing satisfaction and success.

And now, gentlemen, what are the conclusions of this whole matter? These.

Reason and right alike demand a more rational and correct idea as to the origin of the toxic neurosis we have noted.

This demand complied with—regarding such patients, with certain exceptions, as creatures of conditions beyond control, and so no more culpable than the subjects of other functional disorder—will be most helpful against the protective temptation to untruth.

The medico-legal status of such cases will then be more in keeping with advanced forensic medicine.

The medical care of these cases will tend to a more humane method, with a larger promise of good results, both near and remote.

It will, too, be likely to lessen the increase of habitués, and the number now existing, for a more correct idea as to the genesis of this disorder will prompt medical men to greater care in avoiding the cause, while many a patient—who now shrinks from disclosing his misfortune—feeling he is not denied the charity his case deserves, and that he can command resources both helpful and humane, will be impelled to avail himself of the aid that scientific treatment can surely extend.

314 STATE STREET.

A CASE OF SPINA BIFIDA.

BY JAMES W. INGALLS, M.D.

Presented to the Brooklyn Pathological Society, April 12, 1888.

On September 25th, 1888, was called to attend Mrs. H. in confinement. Patient was a primipara about twenty years of age, and a native of Mexico. Both she and her husband were free from any deformity, and had always enjoyed excellent health. Duration of pregnancy about nine months. Upon examination, I found the breech presenting. Labor progressed favorably, and nothing occurred worthy of special note, except that about half an hour before delivery, while making a digital examination, I discovered over the sacrum of the child a loose flap or fold of tissue, the nature of which at that time I was unable to satisfactorily determine.

After delivery I found the following condition: Over the lumbosacral region were two flaps, each two inches and a half long and about an inch wide; the outer borders were free, the middle portions of the inner borders were attached over the spinal column, and at this point of attachment there was an opening which communicated with the spinal canal. This opening was about large enough to admit the tip of the little finger. The anterior surface of the flaps was simply a continuation of the integument, the posterior surface was a continuation and expansion of the membranes of the spinal cord. The edges of these folds were straight and showed no signs whatever of having been torn or lacerated. There was atrophy and complete paralysis of both lower limbs. No other deformities existed. Flaps were placed in close apposition over the opening into the spinal canal, and upon them was put a thick compress, held in place by a wide bandage. The child continued to do well until the morning of the fourth day, when convulsions developed, and death took place in a few hours.

Dissection showed that there was a fissure of the spinal canal extending from the second lumbar vertebra down to the sacrum, the laminæ being absent. Both sciatic nerves were given off in the usual manner.

**THE BROOKLYN MEDICAL
JOURNAL.**

EDITORIAL.

THE FINANCIAL RESULTS OF MEDICAL PRACTICE.

The medical men of the Bay State have been treated several times during the past decade to the mournful story of the meagre financial results from a life-long practice of medicine in that commonwealth. The detailed cases, narrated by Dr. Cotting, were pitiful enough, for they were proof that a faithful, conscientious and skillful medical career could find little laid aside for the "rainy day" of personal illness or the vacation for the tired brain and body, or the reposeful life of a physician's family when death had closed in on his labors. In the same strain Dr. Jeffries, in his late annual address before the Massachusetts Medical Society, proclaims that "no man has made a fortune as a physician, I mean no one ever paid his expenses and laid by at interest enough to live on through the practice of medicine."

This breathes in the atmosphere of complaint as if the profession of medicine were exceptional in life's vocations; as if it, alone of all the lines of work, did not lead to financial results where "enough to live on was laid by at interest." It is very pertinent to ask, in what pursuit in life inheres that tendency to make the laborers therein independent of labor? It is equally pertinent to ask, where is there an instance, in the history of labor, where a man, following the duty common to his fellow workers and relying on his own unaided hands and brain, ever acquired the competency to live, in his accustomed sphere, independent of labor? Dr. Cotting's instances of the poverty of medical men are pitiful, but they are duplicated in the ranks of the promoters of literature, art, science and philanthropic work through historic time and will be multiplied to the end. Great wealth is the possession of but very few and, on the lines of legitimate industry, is always the result of combination and the use made of the labor of others. In the early part of the century, Mr. Astor founded a fortune by buying up pelts from the trappers of the Northwest. Had he depended on what his own hands could have done, his old age would have found him drying his skins and frying his bacon with his own hands in his forest cabin. Mr. Carnegie to-day, utilizing the labor of miners in iron and coal and giving direction to the skill and toil of a multitude of mechanics, is still adding to his fifty millions. Had he depended on the limitations of his own brawn, he might still each evening be washing the grime from his horny hands under the

faucet in the hallway of his tenement house lodgings. These great possibilities of combination are in the genius of commercial enterprise, though they are realized by few. They are foreign to the genius of labor where combination is impossible, and where the labor is of such a character that there is no monopoly of skill and many can accomplish it equally well. A medical man's labor is limited by what he can himself do, personally and unaided. He can neither delegate nor superintend. His income is limited by these personal conditions, modified only by the possession of some exceptional skill and the accidents of popularity or environment. The engrossing character of his occupation hinders him from the experience that justifies outside speculation with acquired capital and restrains him from participation in outside ventures which require freedom both of time and thought. He cannot well add another string to his bow.

The results of combination in trade and the income from professional labor are issues from distinct and opposite sources and have no right to be compared or made the subject of invidious reflection. A number of lawyers, each an expert in a special department, may form a partnership, occupy a common office, each helping the other, the emoluments going to the common fund. This is a sort of combine. But the time is not yet ripe, and probably will never come, for the incorporation of a great Medical Trust, with the names of a specialist in eye, ear, throat, nose, lungs, liver, sphincter ani, corns and fallopian tubes, and so on to the minutest subdivision, with the addition of some general practitioners and apothecaries, displayed around the casings of some common front door, to scoop in the community and pool the receipts on a graded tariff. Trade is essentially selfish and works for the individual. "If you don't work for number one, number two will be working for you." The accumulation of money is neither end nor contingent in professional life. The pursuit and application of medical science are on the higher level with the learning of jurists, scientists, educators and literateurs, whose mission is the unselfish search for knowledge for the immediate benefit of mankind and the advance of civilization.

While it is true that very few in any calling "lay by at interest enough to live on," a very small number of that few do actually retire from active work and live on that interest, and this for two reasons: First, a man in successful professional life is in receipt of an income which enables him to live in luxurious surroundings, gratify tastes and enjoy recreation, which

income, considered as interest, would represent a capital sum exceptional even among the results of successful trade, stock gambling or railroad wrecking. Such a man, and he is one of many, could live on what he “has laid by at interest,” if he saw fit to live in less luxury and sacrifice the gratification of tastes which have been cultivated and become necessary to his comfort. He could live on his interest, but he does not care to live in idleness. On the other hand, the conditions of a cultured life are of an ever widening horizon, and it is characteristic of medical men that their intellectual sense is inquisitive, keen, appreciative and alert in their own sphere of action, less satisfied with what is and more anxious for better results, beyond the genius of any other professional life, and this for the distinctive reason that every new discovery in medical science promotes accuracy in the application of medical art. Working becomes a passion with medical men; the more they know the more eager they are to work. This passion is not to “lay by at interest enough to live on.”

It is quite in the sentiment of medical addresses to bewail the profession as ill-paid, and that, for a learned and self-sacrificing body of men, its labor and accomplishments are very inadequately rewarded. The exact contrary is, probably, very much nearer the truth. There are many learned men in the profession and there is a wide range of special learning which is the common property of the profession, and all are more or less adept in the use of agencies of the art. There is, likewise, a vast amount of patient and uncompensated care given in the routine of practice, which is a natural outcome of the practice of the medical art. It would be absurd to claim the diploma as representing a liberal education or even high special attainments, as it would be ridiculous to assert that a dispensary patient regularly received the attention given to the German Kaiser or General Sheridan. There are instances of failure and poverty among medical men, but when the doctors in the country stand to the population in the proportion of 1 to 580, the assumption is that they have become needful, each to his 580. Doctors have many book charges that are not collected. Laborers are swindled by their bosses, and every business man meets his unlucky customers; the parish gets behind with its rector. The doctor is no worse off than the rest, and besides he has no salary list, and no accommodation at the bank to make good.

Most men are discontented, and the want of contentment is just as querulous with the cosmopolitan reputation that unblushingly pockets a

double eagle for a few raps on the thorax as with its suburban and obscure double that explores a whole chest half an hour for a dollar. The latter pays a shilling to the village blacksmith to reset a shoe, and the former hands over eight dollars every time the farrier looks at his team. Discontent goes with a misfit, and Depew told the Syracuse students that “misfits were everywhere and were always cheap.” It is doubtful if, upon the whole, there are in any walk in life such an unbroken line of splendid fits, the man to his duty and his clientage, as in the medical profession. It is not to be doubted that medical men, each to his location, his culture, his taste and his instincts, are better housed and clothed, more liberally supplied with the machinery of their technique, have greater demands on their purse in the interests of charity and reform which are duly met, have better educated families, have longer and more frequent opportunities for enjoyment which are not wasted, than can be counted item for item on the balance sheet of the average worker in any other profession or occupation. And these are the proofs of financial success, and they put aside the plaint that because the doctors do not “lay by at interest enough to live on” they are an ill-used class of men. The community pays liberally for being taken care of, and it ought to. The medical man’s entire time is taken up in acquiring the experience to exercise prompt judgment in emergencies, and this is precisely what the community pays for and is far from niggardly in the payment. Experience, needful to prompt judgment, is worth more than day’s wages or marginal profits, and this the community recognizes, and its estimate on the value of this experience is generally just. It may not be invariably accurate, but a doctor’s annual cash total is a very liberal estimate of what his individual experience is worth to the community. If the doctor does not “lay by at interest,” it is not because he does not receive enough, but because his relations to life make a free expenditure of money a necessity. He is at a certain disadvantage with a fair share of the people in being compelled to pay his debts. An excellent physician who is also a bohemian or with loose ideas as to honorable obligation, would be a nondescript. He is a fixture in the community with an open reputation, and it is proof of his liberal income that he is able to make and sustain that reputation.

THE OPEN STREET-CAR WHISTLE.

The open street car is in its mid-career for 1888, and the fiend of the whistler is on the vertex of successful practice. The stranglers of the Orient were an occasional incident in that sunburnt civilization as compared with the death-dealing, pestilential prevalence of the Brooklyn open street-car conductor, literally “armed to the teeth” with his offensive weapon, out of whose depths, impelled by æolic volumes from jerky and gigantic costo-diaphragmatic spasms, issue the ear-splitting and nerve-rending combination of fog-horn and prolonged rifle-crack. From stable to terminus and back, circulating along the outer step, holding on to the uprights with extended arms, facing forever the five-cent and helpless “fares,” two to four inches of potential reed or metal protruding from his embracing lips, like an ill-placed proboscis on a witless pachyderm, he summons the driver to screw up his brake and arrest his sportive team for a fare to unload, or to reverse the process for the temporary torture of more victims in hoisting in of other patrons of the line, and the shrill horror of his whistling signal, right in the faces of the passengers, is made more agonizing by the uncertainty of when and on whom it will discharge its blast, being forever ready for action, like the lance in rest of the jousting knight. It would be easy to aim this calliope at the curbstone or the empyrean, but this regard for the passengers’ tympanum disturbeth not the peaceful slumber of the tramway directory, whose shibboleth is the Vanderbilt curse of “the public be ——.” But sadder than the disregard of common-carriers for public comfort is the unearthed conspiracy of the otologists with the ill-paid conductors on the horse-cars. For some years this specialty thrived on the otitis acquired at the bathing-houses at Rockaway and Coney Island, but the public discovered that a little cotton in the meatus was the needful prophylactic, and otitis, as a source of revenue, dwindled to the starvation point. Again, and for a time, the horn of plenty overflows in the otologist’s operating room, and his commissions to the car conductors promise to put them soon on a plane with the diamond-bedecked shirt-fronts of the average hotel clerk. It was said that so possessed was a certain London specialist with the operation of tonsillotomy that these amputated glands were each morning shoveled out of his office by the basketful. There are compensations all through life, and the hordes of cash boys, whose occupation vanished with the introduction of mechanical carriers into the great dry good bazaars, now find ample and continuous employment in sweeping out the heaped up fragments of shattered ear drums from the

infirmaries of otological specialists. Verily, this deal among the ear men with the whistling open car conductors for the embezzlement of the community deserves the most summary and high-handed reprobation. There is but the faintest justification for such combination in the new code, but even that cannot fairly be pleaded when the integrity of the community's ear is imperilled. A proper corps d'esprit would impel to the conservation of a professional brother's prosperity, but even that laudable sentiment must have subordinate place when the profession at large, who are the conservators of society, see that society is likely to turn a deaf ear to the varied forms of human plaint, and all owing to the men who can neither stop or start an open car of a horse railroad without blowing out the ear drums of the community. The public is in peril and who shall be the Curtius to jump into the breach. The conductor cannot be appealed to. He is insensitive, and, besides, he is in authority. One cannot knock the beastly clarion from his lips' embrace: there would be the claim for assault and ejection for disorder. The directors are a weak reed; they dread a strike. Municipal ordinance would be vainly sought: workingmen have a union and votes. The police, even the finest, are not open to bribery: they are at home in a brawl, and noise is their normal condition of repose. The profession must interfere. Henceforth let the cry be "boycott the whistle." If it must exist, let the instruction be boldly posted at the starter's office: "Conductors must aim their whistles at the curbstones and not in the ears of the passengers."

PROMPT TELEPHONE SERVICE.

The telephone is too useful not to be treated properly. It is always an affair of two parties and each is in duty bound to be considerate of the other. The bell rings, it is answered promptly, and patience becomes well nigh exhausted before "central" succeeds in establishing the connection, and the time of the respondent is wasted. The reason for this rests on the thoughtlessness or selfishness of the one who makes the call. He rings and asks for a certain connection, and then hangs up his instrument, goes away to wait for a summons. In the meantime the respondent answers, stays by his instrument, "central" endeavors to call up the caller, perhaps through another office, the connection is often broken, and after much tribulation

the connection is fully made. This is of very frequent occurrence and could be avoided, for the most part, by the caller staying by his instrument for the few seconds usually required to make the connection. There are occasional instances of bad management and some ugliness in the central office, but they are quite rare, and the service is very prompt. More delay and annoyance are caused by thoughtlessness of the users of the telephone than by any neglect of duty on the part of the operators at the central offices. One who is called up has a right to consider that he is wanted, and that promptly. It is the duty of the caller to be careful not to annoy the central office or waste the respondent's time. Moral: When you call, stay by your instrument till the reply comes.

OFFICIAL ORTHOEPEY.

The Mayor has made his appointments to the vacancies in the Board of Education. The proper assumption is that they are all good men and true, able to read, write and cipher. It would be worse than libelous to give houseroom to the rumor that any member of this responsible Board ever "made his mark." One would be properly horrified at the audacity of the narrator of such a tale as the following: A member of a local committee entered the class-room as the teacher was conducting the recitation in spelling from the Reader. After listening for awhile, he intimated his desire "to give out a few words," which desire was politely acceded to, and the book handed to him. A number of words were correctly and promptly spelled, and he gave out the word "Egg-pit." One child after another was downed by the astute member until the teacher, in pity for her flock, suggested that the word was not in the lesson. Smiling disdainfully at her ignorance and presumption, he pointed his No. 11 forefinger to E-g-y-p-t. Tableau. The Directory for 1888 intimates that we live in a city of nearly 800,000 inhabitants.

PROCEEDINGS OF SOCIETIES.

THE MEDICAL SOCIETY OF THE COUNTY OF KINGS.

A regular monthly meeting of the Medical Society of the County of Kings was held at the rooms of the Society, No. 356 Bridge Street, Brooklyn, on June 19, 1888.

The meeting was called to order at 8.30 P. M., with Dr. Wallace in the chair. There were eighty members present.

The minutes of the previous meeting were read, and on motion adopted as read.

The Council reported favorably on the names of the following gentlemen: Drs. Sidney Allen Fox, George H. Treadwell, Fred. L. Goddard, Stanton Allen, Horace B. Scott.

The following gentlemen were declared elected members of the Society: Drs. Chas. S. Fischer, James L. Carney, Eliot Gorton, Heber N. Hooper, Henry H. Morton, Geo. B. Rockwell, Lewis S. Meeker.

The following gentlemen were proposed for membership:

Dr. J. Le Roy Tettemore, 128 Rockaway Avenue; graduated at L. I. C. H., 1881, proposed by Dr. J. H. Hunt and seconded by Dr. W. B. Chase.

Dr. Stanton Allen, 114 Montague Street, graduated at College of Physicians and Surgeons, 1881, proposed by Dr. Richmond Lennox and seconded by Dr. J. S. Prout.

Dr. James W. E. Roby, 115 Lee Avenue, graduated at Medical Department, N. Y. University, 1887, proposed by Dr. D. Myerle, seconded by Dr. W. M. Hutchinson.

Dr. Charles G. Purdy, 56 Pulaski Street, graduated at the University of the City of New York, proposed by Dr. Chase and seconded by Dr. Little.

The Secretary stated that the name of Dr. Stanton Allen was among the propositions for membership presented at this meeting, and also among those reported favorably by the Council.

He said in explanation that Dr. Allen's diploma had been sent to the Board of Censors and passed upon before his name had been proposed, in order that the diploma might not remain with the Censors till next September.

This was an infringement of the By-Laws upon admission of members, but if no objection was raised, the By-Laws would be waived in this case.

No objection was offered.

SCIENTIFIC BUSINESS.

The first paper of the evening, "On the Relation of the Bacillus Tuberculosis in Pulmonary Phthisis," was read by Dr. I. H. Platt, of Lakewood, N. J., and discussed by Drs. G. R. Butler, J. M. Van Cott, J. H. H. Burge, P. H. Kretzschmar and G. A. Evans.

The next paper was "A Case of Dystocia and Double Phlegmasia-dolens," by Dr. Lucy M. Hall. This was discussed by Drs. Dickinson, Thayer, Skene, Chase, Schenck and Harrigan.

A paper, entitled "Note on the Disinfection of Physicians' Clothing," by Dr. R. L. Dickinson, was then read and discussed.

REPORTS OF COMMITTEES.

The President called for the report of the Obituary Committees upon the late Drs. Chapman and Mitchell.

The obituary report of the late Dr. Chauncey L. Mitchell was then read by Dr. Burge, as follows:

Chauncey L. Mitchell, whose ancestors were of Puritan stock, coming to this country from Halifax, England, as early as 1635, was born in New Canaan, Connecticut, November 20, 1813. An excellent general education, obtained here and at Union College, Schenectady, was supplemented by a full course of study in the medical department of the University of the State of New York, better known as the College of Physicians and Surgeons, where he was graduated in 1836. The next twelve months were spent in the New York Hospital, and the succeeding two years on the continent of Europe. All who knew Dr. Mitchell intimately are so familiar with his studious habits that they need not be told that he never lost an opportunity for observation and improvement. This was true of him, not only during the period of his pupilage, but in all the fifty years of his active practice, five of

which were in the City of New York. He came to Brooklyn in 1844, was admitted to membership in the Society of the County of Kings soon after, and was thrice honored by an election to its highest office. All the duties of his active life he performed with dignity and zeal. No one more than he enjoyed the uninterrupted confidence and affection of this community. Among his friends and patients were numbered the best of our citizens, and he is now equally missed in the profession, in the church and in the household. Dr. Mitchell was an earnest and devout believer in the Christian religion. His connection with the Church of the Pilgrims antedates the pastorate of the Rev. Dr. Storres, between whom and himself there was an intimate, personal and professional relationship for forty years.

Dr. Mitchell's powers of observation and discrimination and his exactness of verbal expression were so excellent, that we regret that he did not give more time to authorship. An article on "The Effects of Ergot," "Labor Complicated with Disease of the Heart," and an occasional contribution to the journals, are all that we can find of his writings.

During the period of declining health, which occupied more than two years, Dr. Mitchell resigned many positions of responsibility, yet, at the time of his death, he was Sn. member of the Council of the Long Island College Hospital, member of the Amer. Med. Asso., Consulting Surgeon to St. John's Hospital, to L. I. Coll. Hospital and to the Home for Aged Men, a member of the New York Academy of Medicine, New York Co. Med. Soc., and Kings Co. Med. Asso. He was also a life member and Corres. Sec't'y of the L. I. Historical Society. It is matter of record that he was once a member of the Medical Staff of the Brooklyn City Hospital, and that he also filled with honor the Professorship of Obstetrics in the Castleton Medical College.

In 1843, Dr. Mitchell married *Caroline*, daughter of Hon. B. F. Langdon; in 1857, *Frances*, daughter of Hon. Benjamin Wright; in 1875, *Kate*, daughter of Hon. J. M. Van Cott, of this city.

Dr. Mitchell's tastes were professional, literary, artistic and religious, but the centre of all was his own home. In the house which he had built more than forty years before, he died on the 8th of May, 1888, terminating a long career of untiring usefulness.

Mr. Chairman: In the preparation of this minute for the records of the Society, your committee have recognized the fact that they were appointed

for this simple duty only. The pronouncing of a suitable eulogy belongs to other hands and to another occasion. We offer for your consideration the following:

Whereas, In obedience to the Divine Law the fully matured life of Chauncey L. Mitchell is ended on earth, and

Whereas, He was destined to fill a high position in the profession of medicine and as a citizen of this country, and

Whereas, He met all these requirements, fulfilled every duty, and discharged every obligation in such manner as only a highly cultivated, educated and honest man could, and

Whereas, When the end came, those who knew him best could truly say that he had left nothing undone to complete a noble, highly useful and honorable life;

Therefore be it resolved, That while paying this tribute to his memory we desire to express our high appreciation of his many virtues, and that while we deeply feel our loss, we also cherish his memory, and are grateful for the honor, dignity and advancement which his life's work gave to this Society.

Resolved, That we offer our deepest sympathy to his bereaved family, and that a copy of these resolutions be conveyed to them as a humble tribute to his superior worth.

All of which is respectfully submitted by your committee.

J. H. HOBART BURGE, M.D.,
ALEX. J. C. SKENE, M.D.

The report of the Obituary Committee, as above, was accepted and committee discharged.

The Resolutions introduced by this committee were adopted as read.

NEW BUSINESS.

The Chairman read a communication from the Secretary of the Kings County Pharmaceutical Society, stating that the term of office of the two members of the Board of Pharmacy from this Society had expired, and asking that their places be filled.

THE PRESIDENT.—As I understand it, our elections take place only at the end of the year, and if this communication is to be acted upon, it will be necessary for the Society to pass a special resolution authorizing the election of these gentlemen. If it is the desire that the Society pass such a resolution, a motion will be in order.

A MEMBER.—I move that the By-Laws be suspended so that the election may be had this evening. Carried.

THE PRESIDENT.—Nominations are now in order. The present incumbents are Dr. J. H. Hunt and Dr. C. E. De La Vergne.

A MEMBER.—I move that the two present incumbents be continued in office, if it be the voice of this Society, until the annual meeting.

Seconded and carried.

There being no further business, the Society adjourned.

W. M. HUTCHINSON, M.D., *Secretary*.

PROGRESS IN MEDICINE.

PREVENTIVE MEDICINE.

BY ELIAS H. BARTLEY, M.D.,

Professor of Chemistry and Toxicology, and Lecturer on Diseases of
Children, Long Island College Hospital, Brooklyn.

THE GERM THEORY A CENTURY AGO.

Under this caption the British Med. Journal for February 11, 1888, contains an editorial review of a pamphlet of 87 pages, published in 1788, and entitled: "*A Treatise on Fevers, wherein their Causes are exhibited in a new point of view, to prevent Contagion; and Putrid Sore Throat, Inflammatory Fluxes, Influenza, Consumptions, as well as the Low Nervous Fevers that terribly affect the Spirits, may be cured with ease.*"

The most remarkable part of the book is the speculative or explanatory part, consisting of an exceedingly ingenious argument, based upon the analogy of admitted facts, to prove that the cause of contagious fevers is some invisible noxious matter in the air. Of the intimate nature of this matter he says: Some consider it to be a sulphurous exhalation from the earth; but this cannot be, for, if so, acrid and sulphurous fumes would increase it, instead of checking or annihilating it. Another theory is that it is due to the products of putrefaction; but how can dead putrid matter ever get such activity as to work such astonishing results? It must therefore be something endowed with a more powerful activity than anything belonging to the mineral kingdom or simply putrefying matter, and must, therefore, be something "actually living." He further concludes that these living organisms must have an existence independent of the body in which they are found. For this view, surprising and novel enough at first, loses some of its singularity, if we search for resemblances elsewhere. Now, just as it was well known that itch is due to the presence of acari, insects visible by the aid of the microscope, so close attention to these matters in numberless cases during many years, has proved beyond a doubt that the gaol distemper, putrid fever, plague, and infectious epidemics generally, proceed

not from matter putrid in itself, but from invisible insects also, that, floating in the air at times, are lodged in the skin in immense quantities; feeding here in clusters, they produce pimples, pustules, etc.; for instance, the eruption of small-pox. He overlooks, or fails to mention, the possibility of their entering by the air passages or digestive system. "Medicines," he says, "which poison insects without injuring the constitution have always proved specific." These insects, which constitute contagion, are communicated by air, the raiment, as by contact. He admits that vegetables as well as animals suffer from the ravages of these animalcules. He believed that they originated from eggs and not *de novo*. He advises fumigations with sulphur and frankincense to destroy contagion in rooms, and shows that many diseases in lower animals are cured or prevented by the use of certain agents known to kill insects.

In summing up his theory, he says that, generally speaking, there are two sources of these animalcules. First, from subterranean sources, which operate in all sorts of weather and are accompanied by electrical phenomena. Second, from the surface of the earth, swamps, filthy lakes, stagnant ponds, etc. The eggs left on the soil develop in summer, and "the multitudes effluvia into air."

The essay is interesting to us because of the very clear foreshowing of a theory that we are apt to regard as the creation of recent years. It is a good example of the power of attentive observation and inductive reasoning, which is so seldom met with even in scientific medical men of the present day.

ALBUMINURIA A FREQUENT RESULT OF SEWAGE POISONING.

Dr. George Johnson, in *Br. Med. Jour.* for March 3d, gives the histories of four cases of albuminuria which he believes were the result of breathing sewer air. In addition to other diseases, the result of drain poison, the author has met with several cases of albuminuria which he believes can and does under continued exposure to the sewer poison, result in incurable disorganization of the kidneys. He thinks that, in the absence of other probable exciting causes of albuminuria, the possibility of sewer poisoning should be constantly borne in mind. It is needless to dilate upon the importance of discovering the exciting cause of a disease so serious in its

consequences as nephritis. In each of the four cases cited, albuminuria and casts were found in the urine, and blood in two of them. In each case defective drainage was proven, and in two of the four an immediate improvement occurred on removing this cause. One proved fatal from suppression of the urine.

The author suggests as an interesting point, that amongst the various diseases resulting from drain poison, diphtheria is in a very large proportion of cases associated with albuminuria.

It would be interesting to know whether a large proportion of cases of diphtheria occurring in houses having defective plumbing suffer with albuminuria, than in houses where no such defect exists. If these observations are confirmed, we may learn from them something of the cause of the great fatality of scarlet fever and diphtheria in houses which contain defective drains.

SEWER-AIR POISONING.

The question of sewer-air poisoning has received no inconsiderable attention at the hands of sanitarians within the past few years, some claiming that it is a carrier of many of the contagious diseases, including malarial affections, while others have denied its harmful action in these respects.

The last class, in substantiation of their claim, point to the assumed fact that plumbers and those who work in sewers are not, as a rule, especially subject to the diseases generally attributed to sewer air. That plumbers are not exempt from troubles of this kind is attested by numerous examples. According to *Science*, an inquest was recently held in Liverpool, Eng., on the body of a plumber's apprentice who had been engaged in repairing pipes which connected with the sewer. Quantities of gas came through these pipes, and at the time the young man complained of pain and sickness, and died forty hours afterward. The jury rendered a verdict of poisoning by sewer air.

According to the *Sanitary News*, Dr. Vaughn, of the Michigan State Laboratory of Hygiene, claims to have found the specific germ of typhoid

fever in the air of a soil pipe from the prison at Jackson, in that State, during an outbreak of typhoid fever.

The *Sanitary Inspector* for February and March, reproduces from the *Medical News* an article by Dr. Henry Hun upon this subject. Dr. Hun cites twenty-nine cases in support of his statements. The histories are those of non-contagious diseases, and therefore were probably cases of illness produced by non-infected sewers. He says: "In all of these twenty-nine cases there was an escape of a large amount of sewer gas into the air which the patients breathed; and at the time that the case was observed, it seemed extremely probable that the sewer gas was the cause of the disease.

"From the consideration of these twenty-nine cases, we may conclude that it is probable that the following conditions may result from sewer-air poisoning:

"1. Vomiting and purging, either separately or combined.

"2. A form of nephritis.

"3. General debility, in some cases of which the heart is especially involved.

"4. Fever, which is frequently accompanied by chills.

"5. Sore throat, which is frequently of a diphtheritic character.

"6. Neuralgia.

"These conditions may occur separately, but are frequently combined, and it is especially common for the fever to be associated with other symptoms of sewer-gas poisoning. Finally, in cases of sewer-gas poisoning, there is one group of symptoms which is almost always prominent, and these symptoms are: loss of appetite, drowsiness, extreme prostration, and a dull, unpleasant feeling in the head; and whenever this group of symptoms occurs, not as the result of an attack of acute disease, but as a chronic condition, we are justified in suspecting that the patient is exposed to sewer-gas infection."

EFFECTS OF FOOD PRESERVATIVES ON THE ACTION OF DIASTASE, PANCREATIC EXTRACT AND PEPSINE.

This subject has recently received experimental study at the hands of Dr. Henry Leffman and William Beam, the results being published in the *Analyst* for June, 1888.

The antiseptics selected were those which have been known to be used to preserve articles of food and drink. They were salicylic acid, boric acid, sodium acid sulphite (sodium bisulphite), saccharine, beta-naphthol and alcohol.

In the following experiments a solution of arrow root starch, 30 grains to the litre, was used.

To 100 c.c. of this solution was added 0.5 c.c. of maltine diluted to 50 c.c. with water.

The figures give the proportion of antiseptic to the whole volume of liquid.

Experiments with Maltine.

<i>Antiseptic used.</i>	<i>Amount.</i>	<i>Fehling's Solution reduced by the Maltose formed.</i>
None.	None.	245 cc.
Salicylic acid.	1 to 500.	No sugar formed.
Salicylic acid.	1 to 1,000.	No sugar formed.
Salicylic acid.	1 to 20,000.	245 cc.
Boric acid.	1 to 1,000.	245 cc.
Sodium bisulphite.	1 to 1,000.	245 cc.
Saccharine.	1 to 1,000.	18.5 cc.
Saccharine.	1 to 500.	5.6 cc.
Beta-Naphthol.	1 to 1,000.	204 cc.
Beta-Naphthol.	1 to 500.	174 cc.
Alcohol.	1 to 25.	245 cc.

Experiments with varying amounts of diastase showed that one part of salicylic acid to 1,000 of liquid prevented the diastasic action completely. Saccharine in the proportion of 1 to 1,000 prevented the formation of sugar when the proportion of diastase was 1 to 1,000 of liquid. When the proportion of diastase was reduced to 1 in 2,000, salicylic in the proportion of 1 to 3,000 prevented the formation of sugar. It seems, then, that the weaker the diastasic solution, the more is its action hindered by salicylic

acid, saccharine, etc. Sodium bisulphite has little if any power of hindering diastasic power.

Experiments with Fairchild's Pancreatic Extract.

<i>Antiseptic.</i>	<i>Amount.</i>	<i>Fehling's Solution required.</i>
None.	None.	78 cc.
Salicylic acid.	1 to 1,000.	No sugar formed.
Saccharine.	1 to 1,000.	No sugar formed.
Beta-Naphthol.	1 to 1,000.	78 cc.
Boric acid.	1 to 1,000.	78 cc.
Sodium bisulphite.	1 to 1,000.	80 cc.

The tests were made with 0.2 grams of the extract.

It seems from these experiments that salicylic acid and saccharine, in the proportions used, entirely prevent the action of pancreatic ferment upon starch.

In similar experiments with saccharated pepsine with hydrochloric acid, except that the temperature was kept at 105° F., sodium bisulphite and boric acid were without effect.

Saccharine and salicylic acid had a slightly retarding action. Beta-naphthol almost entirely prevented the action.

With pancreatic digestion of albumen the results were practically the same, but the retarding action of the salicylic acid and saccharine was not quite so well marked.

From these experiments it will be seen that salicylic acid prevents the conversion of starch into sugar under the influence of either diastase or pancreatic extract, but does not seriously interfere with peptic or pancreatic digestion of albumen. Saccharine holds about the same relation as salicylic acid.

Sodium bisulphite and boric acid are practically without retarding effect.

Beta-naphthol decidedly interferes with the formation of sugar by diastase, but not with the action of pancreatic extract on starch.

It almost entirely prevents both peptic and pancreatic digestion of albumen.

The bearing of these experiments upon the sanitary question of permitting the use of these preservatives in foods, is self-evident. Prof. Leffman says: "Their use is scarcely allowable under any circumstances, and certainly only when the nature of the preservative and the amount is distinctly stated." The use of saccharine as a sweetening agent must be looked upon as deleterious to health, and ought to be forbidden by sanitary authorities.

MILK AND SCARLATINA.

In a recent number of the JOURNAL, we published an abstract of the reported investigations of Mr. Power, Dr. Cameron and Dr. Klein of a disease among cows, which they believed had caused scarlatina among persons using the milk. The conclusions reached by these gentlemen seemed so startling that the Agricultural Department of the Privy Council began an investigation of the disease. The investigation was given into the hands of Dr. Cruikshank, whose reports are published in the *British Medical Journal* of December 17, 1887, and January 21, 1888.

We have only space here to reproduce the conclusions reached by Dr. Cruikshank, which are as follows:

1. The nature of the contagium of scarlet fever is unknown.
2. The micro-organism regarded by Dr. Klein as this contagium is the *streptococcus pyogenes*.
3. *Streptococcus pyogenes* is found sometimes in company with *staphylococcus pyogenes aureus*, as a secondary result in scarlet fever and many other diseases.
4. A streptococcus was first observed in scarlet fever by Crooke, later by Löffler, Huebner and Bahrdt; but its exact relation to scarlatina, and its undoubted identity with the streptococcus from pus and puerperal fever, was definitely established in 1885 by Frankel and Freudenberg.
5. Both the Wiltshire and Hendon cow diseases were called cow-pox by the people on the farms.
6. Both diseases correspond in their clinical history.

7. The ulcers on the teats correspond in naked eye and microscopical appearances, and the latter vividly recall the appearances of cow-pox.

8. Calves inoculated from the discharges of the ulcers are similarly affected.

9. *Post-mortem* examination of such calves, or of calves inoculated with streptococci isolated from scarlet fever cases, show similar appearances.

10. The *post-mortem* appearances in such inoculated calves are the result of septicæmia.

11. There are no specific visceral changes in cow-pox, apart from complications or coincident affections.

To the above criticism of Dr. Klein's investigation of the Hendon cow disease, this gentleman claims that Dr. Cruikshank studied a different affection, and that the organisms were not the same. It is insisted upon by Dr. Klein, that Dr. Cruikshank's conclusions were the result of studying cow-pox and not the peculiar disease he described as the Hendon cow disease.

THE CAUSE OF DEATH IN PHTHISIS.

Dr. R. W. Philip has made an experimental study to determine the cause of death in phthisis. The results of his study are published in the *Brit. Med. Jour.* of Jan. 28th, 1888.

His experiments were conducted with an extract prepared from fresh sputum from phthisical patients, as follows:

The sputum was treated with alcohol, put in a sterilizer, and heated to 37 to 40° C. for some time, filtered clear, and evaporated at a low temperature until the alcohol was expelled. This extract was used for subcutaneous injections in frogs, mice, and rabbits.

This extract was found to possess very marked toxic properties upon these animals, which manifested themselves by a depression of the higher nerve centres and of the heart. The depressant action upon the heart seemed to be exerted through the cardio-inhibitory mechanism, and is more or less completely antagonized by atropine.

The toxic principle he believes to be the result of the growth of the tubercular bacilli, and allied to the ptomaines. He found the quantity of the substance to be extracted from the sputum to be proportional to the abundance of the bacilli present in it. These observations are in accordance with the observations of various experimenters with other pathogenic organisms, and with the theory that seems to be gaining ground that immunity is the result of such by-products of the growth of these organisms.

SURGERY.

BY GEORGE R. FOWLER, M. D.,

Surgeon to St. Mary's Hospital and to the Methodist Episcopal Hospital,
Brooklyn.

CONTRIBUTIONS TO THE STUDY OF MYXŒDEMA FOLLOWING TOTAL OR PARTIAL EXTIRPATION OF THE THYROID BODY.

J. L. Reverdin (Congrès Français de Chirurgie, 2 session, Paris, 1886.) This disease, following frequently in the after-history of cases of extirpation of the thyroid body, and called by Kocher, of Berne, cachexia stumpriva, was first described by Reverdin, and by him called "operative myxœdema." His description of this disease coincides in general with the views now generally held, although we find some not unimportant deviations from Kocher's conclusions; for instance, the latter observed the disease twenty-four times following thirty-four operations, it appearing to attack by preference those who had not attained their full development. On the other hand, R., basing his experience upon copious statistics, believes the disease is comparatively rare, following upon operations of the thyroid in only twenty-seven per cent. of cases. In his experience, it likewise preferably attacks children and young persons, but that it occasionally fails to occur after complete thyrotomy. Further, R. has observed a milder form of the disease, differing essentially from the graver type of the affection, which latter invariably tends progressively to a fatal termination. In the milder form, the disease is described as oscillating between relapses and improvement, lasting for years, cure sometimes resulting. Several cases are detailed supporting these observations. In two of these, it was supposed that the thyroid had been extirpated in toto, but it was subsequently discovered that a small portion of the gland had remained. Three cases are worthy of especial note. In these, after partial extirpation, in one the right lobe, and in two the left having been removed, an imperfect form of the disease made its

appearance after several months. The characteristic signs, such as swelling of the face and limbs and hesitancy of speech and of muscular movements were absent; while the other symptoms, such as general weakness, pains in the limbs, chest, and head, greater or less loss of memory, chilly sensations, reminded one distinctly of the more complete form of the disease. In all these cases slow improvement followed. Another case is worthy of notice from the fact that, two months after the operation, the remaining lobe had so atrophied as to be scarcely distinguishable. Cases of this atrophy following partial extirpation have been observed both in Germany and in England.

UPON LUXATION OF THE HEAD OF THE RADIUS FORWARD.

Raestock (*Deutsche Militärärztl. Zeitschrift*, 1887, p. 331) has, by means of experiments upon the cadaver, shown that this accident occurs in forced pronation while the radius is resting upon the ulna at the point where the former crosses the latter, the ulna acting as a fulcrum. The head of the radius is pressed against its capsule and tears the latter. More rarely, the accident may occur in forced supination, by pressing the bone against the posterior edge of the ulna, the head of the radius being luxated through a rent made by a rupture of the external lateral ligament between the outer and inner sides; upon pronating the hand, this outward dislocation is converted into a forward one. The annular ligament is either torn or else the coronoid process of the ulna is broken. In either case the interposition of the annular ligament in the fold of the joint becomes an obstacle to reduction. In the experiments as detailed, great force was necessary in the production of the luxation.

Should active and passive movements, manipulation, etc., fail to remove the obstacles to reduction, the author advises a resort to secondary resection of the displaced head of the radius; this, with proper precautions, is certainly a most wise and rational procedure.

CHOLECYSTOTOMY, WITH LIGATURE OF THE CYSTIC DUCT.

Zielewicz (*Centralblatt f. Chirurgie*, No. 13, 1888,) proposes in addition to the so-called “ideal” method of Spencer Wells, that of suturing the gall

bladder to the abdominal wound, to ligate the cystic duct, in order to insure the patient against the return of the biliary lithiasis. The only case in which he has tried it was one in which an attempt was being made to perform cholecystectomy, but the adhesions between the gall bladder and liver were so dense and unyielding as to render the removal impossible, on account of severe hæmorrhage. He therefore passed two ligatures around the cystic duct and divided the latter between them. The gall bladder was then fastened to the abdominal wound, incised and emptied of its contained calculus and biliary secretion. The patient made a good recovery, a fistula remaining, of which the writer says, that “after a time it was almost closed.”

The author claims for this operation the following advantages: 1st. Rapid healing without a resulting fistula. The gall bladder is practically removed from the organism. With appropriate after treatment, its secretion soon ceases, and it becomes obliterated. 2d. The operation is simple and less dangerous than cholecystectomy, and gives the same results.

In contrasting this operation with cholecystotomy, it may be said that the latter simply aims at removing the existing calculi, and makes no provision against the recurrence of the same. Where the “natural” method of Bernay is adopted, and the gall bladder dropped back into the abdominal cavity after suturing the incision made in its walls for the removal of its contents, in case of a recurrence of the disease, the entire operation must be repeated. In the “ideal” method of Spencer Wells, only an incision need be made, in such an emergency, at the site of the first operation. Troublesome fistulæ, however, are apt to remain.

In cholecystectomy, on the other hand, hæmorrhage from breaking down of adhesions between the gall bladder and the surface of the liver, it is claimed, is a frequent and troublesome complication. It is claimed by Thiriar, however (“De l’intervention chirurgicale dans certains cas de lithiase biliaire,” *Revue de chirurgie*, 1886, No. 3), that cholecystectomy is a less dangerous operation than simple cholecystotomy. Again, by Bardenheuer, that hæmorrhage from the liver can be readily arrested. The abstractor witnessed an operation in which the liver was accidentally wounded and the resulting hæmorrhage arrested by the thermo-cautery.

Hertin, a French naval surgeon, in 1767, after experiments made upon dogs, proposed, in wounds of the gall bladder, extirpation of the latter, after ligation of the cystic duct. In these experiments he demonstrated the

feasibility of the operation of cholecystectomy upon the lower animals, at least.

Campaignac, in 1826, proposed ligature of the cystic duct, with partial resection of the gall bladder (Journ. hebdom. Bd. ii, 1829). K. Zagorski has recently attempted this latter procedure on dogs, with fatal results (Przegl. lekarski, 1887, No. 48). Nevertheless, to Zielewicz belongs the credit of being the first to demonstrate, by its successful performance, the feasibility of combining in man the two operations of ligature of the cystic duct and cholecystotomy with suture of the gall bladder to the abdominal wound. Upon further trial the operation may prove to be not only feasible, but to follow out a rational indication with relative safety.

SUPRA-PUBIC PROSTATECTOMY.

A. F. McGill, F.R.C.S. (*The Lancet*, February 4, 1888). The operation consists of two parts: (1) The opening and drainage of the bladder; and (2) The removal of the prostatic valve which prevents the egress of the urine. A full sized silver catheter, curved according to the nature of the case, is passed into the bladder, its contained urine withdrawn and its cavity washed out with a warm saturated solution of boracic acid till this is returned clean and unchanged. The usual rubber rectal bag is now introduced and filled with fourteen ounces of water. The bladder is now rendered prominent by injecting it with a sufficient amount of warm boracic acid solution. The catheter is retained in the bladder, and the fluid with which the latter has been distended, prevented from escaping. The usual median supra-pubic incision is now made, the bladder exposed and made to project into the abdominal wound by depressing the catheter. A large curved tenaculum is now passed transversely into the bladder, touching as it goes the point of the catheter. An incision is now made longitudinally through the bladder wall, the fluid being prevented from escaping by plugging the opening with the finger. The bladder is now seized with nibbed forceps, and applied on each side of the incision, the catheter is withdrawn from the urethra and the bag from the rectum, and the first part of the operation is complete. The interior of the bladder and its neck is now examined, in order to ascertain the exact nature of the prostate enlargement. A pedunculated middle lobe can be removed with the curved scissors, but in the case of a sessile middle lobe,

this must be assisted with the finger and forceps. The “collar” enlargement is removed with greater difficulty. In order to insure the patency of the urethra, it is advised to pass the forefinger into the canal as far as the first joint. It is claimed that the hæmorrhage is not excessive. The operation completed, drainage is effected by passing a rubber tube out of the abdominal wound, the latter being partially closed by a point or two of suture. The tube is removed after forty-eight hours.

The author’s experience is limited to five cases, four of which have proved successful, while the fifth case is still under treatment. He claims that, in cases operated upon early, before diseased bladder or surgical kidney have developed, a radical cure may confidently be expected.

Two objections to this method occur to us: (1) Whoever has performed or witnessed supra-pubic cystotomy, either for the purposes of removal of a calculus or a neoplasm from the interior of the bladder, must have been struck by the difficulties in the way of a thorough appreciation of the condition of its posterior wall low down, or of the cystic neck. Unless specially devised instruments are available for each particular form of prostatic enlargement, it would seem to be a matter of great uncertainty as to just how much of the growth is removed. (2) Until satisfactory granulation of the wound surfaces has been accomplished, drainage, to be efficient, must be facilitated by placing the patient upon one or the other side, a position difficult to maintain, particularly in old people.

RESEARCHES UPON THE VAGINAL PROCESS OF THE PERITONEUM AS A PREDISPOSING CAUSE OF TENDENCY TO EXTERNAL INGUINAL HERNIA.

H. Sachs (Archiv. f. Klinisch. Chirurgie, Band xxxv., p. 321–372) advocates quite decidedly the view, basing his opinions upon preparations of the spermatic cord examined and upon microscopic examinations of cross sections of the latter, particularly as to the relations of the vas deferens and the vessels to the vaginal process of the peritoneum, that the latter is formed before the beginning of the descent of the testicle rather than as a portion of the abdominal wall formed or dragged into position by the testicle in its descent. In proof, he alleges that he has always found, in cases of incomplete descent of the testicle, that organ upon the posterior wall of

the vaginal process, and not on the floor of the same. In females, the formation of the canal of Nuck cannot be said to be due to any dragging.

The entrance to the opening of the vaginal process is found covered by a valve arrangement, and the same is particularly noticeable in the canal of Nuck. The opening of the vaginal process can be caused to gape through a spreading out of the mesentery attached to the ilium or that of the sigmoid flexure. The diameter of the opening is, in general, greater on the right side than on the left in boys, while in girls this difference is not observed. Further, the different forms of the incompletely obliterated opening of the vaginal process agrees with the most frequently occurring forms of the hernial sac in inguinal hernia. The relations of the vaginal process to the elementary parts of the spermatic cord are not constant. On the contrary, the relations which the smooth muscular structures of the cord bear to the vaginal process, in so far as their arrangement into bundles, and their positive relation to the posterior and lateral walls of the same are concerned, are quite constant, and almost form an integral part of the same. The obliteration of the vaginal process depends upon a granulating process, which begins in the middle third of the funicular portion, and from thence proceeds more rapidly in a downward than in an upward direction. This granulation formation takes place essentially during the first ten to twenty days after birth; after this time it takes place more slowly. The canal of Nuck, on the contrary, is found to have almost entirely disappeared at the time of birth. They are both found to be more frequently open upon the right side.

From these observations it would appear that it is not essential to the production of inguinal hernia that a broad and short inguinal canal should be present. The only essential predisposing cause, in children at least, depends upon the condition of the vaginal process of the peritoneum itself.

The question of the legal responsibility of employers is an interesting one, in connection with this question. Hernia cannot be considered as an accident, in the surgical sense, according to Socin (*Korrespondenzblatt f. Schweizer Aertze*, 1887, No. 18), but is really a slowly occurring disease, to which certain well-defined anatomical peculiarities act as predisposing causes.

ACID SUBLIMATE SOLUTION IN SURGERY.

E. Laplace (*Deutsche Med. Wochenschrift*, No. 40, 1887), after repeated and careful examinations and experiments, became convinced that dressing materials consisting of wood-wool, made with sublimate in the usual manner, were far from being germless themselves, much less efficient as antiseptic applications. Gauze, however, showed much better results, but were far from realizing an idealistic asepsis. He likewise found that ordinary sublimate, in the presence of albuminous material, is quickly precipitated and becomes at once ineffective. L. experimented at first with hydrochloric acid as a means of preventing changes in the sublimate from occurring in the presence of organic matter, and particularly albuminous material. But, as hydrochloric acid itself was far from possessing the stability needful for the purpose of preparing dressings, he substituted for it, with the most gratifying results, tartaric acid. The proportions are as follows: sublimate, 1 part; tartaric acid, 5 parts; distilled water, 1,000 parts.

OPERATIVE TREATMENT OF PYOTHORAX.

E. Rochelt (*Wiener med. Presse*, No. 32 and 38, 1887). The expansion of the lung is greatly impeded after the usual operation for empyema by incision, by the free entrance of air in the pleural cavity. Mader, Subbolik and Immerman devised means for preventing this. R. operates by first resecting a rib, leaving the periosteum intact, and subsequently opening the pleural cavity by means of a trocar and canula. A drainage tube accurately filling the latter is now introduced, through which a disinfecting fluid is injected and its outer opening closed by means of a spring clamp. The tube is connected to an aspirating bottle, into which the pus is discharged. The tube is again clamped, and the bottle into which the pus has been aspirated removed, being replaced by another containing a sublimate solution, 1 to 500. Removal of the clamp and raising and lowering the bottle thoroughly irrigates the pleural cavity. This being accomplished, the patient holding his breath in expiration, and the clamp again applied, the irrigating bottle is removed, and a short hard rubber tube connected to the outer end of the drainage tube. This hard rubber tube has a soft rubber diaphragm which acts as a check valve, effectually preventing the ingress of air during inspiration, but in no wise interfering with the egress of fluid from the pleural cavity during expiration, fits of coughing, etc. For purposes of further irrigation

the short rubber tube containing the valve may be removed after guarding against the entrance of air by clamping the drainage tube beyond, and the washing bottle reapplied. During the intervals of irrigation, absorbent antiseptic dressing are kept applied.

The abstractor would suggest the application of this method, particularly in recent cases and in children, without the previous resection of a rib. The increased support afforded the canula by the greater thickness of the thoracic walls would be a still greater safeguard against the entrance of air into the pleural cavity. A certain proportion of acute cases will recover without resection of a rib.

WOUND-HEALING UNDER THE DRY ASEPTIC SCAB.

Prof. Kuester (Centralblatt f. Chirurgie, March 17, 1888,) in reply to remarks made by Dr. Sonnenberg before the Association of Berlin Surgeons, January 9, who characterized K's method of treating the wound after the operation for the radical cure of hernia as an "open method," objects to this designation of his method, and takes occasion to more fully describe his method as follows: After the reduction of the contents of the sac, the latter is sutured and excised, and the ring is also closed by suturing. The wound cavity itself is now closed by several rows of buried sutures, so arranged as to bring together the edges of the several layers, tissue to like tissue. In congenital cases he does not loosen the sac, but sutures its opposing surfaces down to the point where the testicle lies free. The wound of the skin is now closed by a continuous silk suture, and an iodoform and collodion mixture brushed over the line of suturing until it is perfectly covered in and blood no longer oozes through. No drain is used, and no further dressing is deemed necessary. If, after two or three days, a split occurs in the scab or crust formed by the drying of the iodoform and collodion, the gap is quickly filled by a slight oozing from the deeper portions of the wound, which, upon drying, becomes a bar against infection.

There can be no doubt, if thorough asepsis is observed and obtained during the operation, the method of completely obliterating every space in which blood clot or serum could accumulate would do away with the necessity for drainage. This granted, it follows, as a natural sequence, that absorbent dressings are superfluous, simple protection of the line of

suturing from atmospheric influences, infection, etc., being alone indicated. The iodoform and collodion compound would seem to fulfil this admirably. The method could scarcely find application in large or deep wounds, particularly if the latter invaded planes of dense connective tissue, fasciæ, etc. Here it would be manifestly best to provide drainage, etc.

In marked contrast to K's method is that of McBurney, who, providing against infiltration by suturing the entire thickness of each edge of the wound together in such a manner as to render it practically but one layer, packs the wound cavity, and thus obliterates the inguinal canal, the latter filling up by granulation, a firm cicatricial plug taking its place.

THE TREATMENT OF CAROTID HÆMORRHAGE.

Mr. Frederick Treves (*The Lancet*, January 21, 1888). In the neck, pressure upon the carotid artery, in hæmorrhages from the branches of that vessel, cannot be applied in the ordinary way with success, nor could it be maintained for a sufficient length of time, if the pressure succeeded in arresting the hæmorrhage, to be of service. Treves proposes, however, to occlude the vessel temporarily by throwing a broad piece of catgut around it, tying it in a loose loop, and then making traction upon the same. The circulation through the vessel is at once arrested, but can be at once restored upon relaxing the tension upon the loop. He relates four cases in which the method proved successful, so far as the arrest of the bleeding was concerned. One of the patients succumbed to the great loss of blood sustained prior to the application of the ligature, although the other carotid had been previously tied. In the first case the loop remained *in situ* for four days. The second case was the fatal one. In the third and fourth cases the loop was removed on the seventh day.

The method is based upon the fact that temporary arrest of the circulation in certain cases of hæmorrhage from the limbs, where ordinary means may be employed to exercise pressure, are quite sufficient, not only for the purposes of a temporary expedient, but also seems as a curative measure. This temporary modification of the blood current may be all that is required in many instances. Just how long the blood current may need to be checked, must be carefully studied in each individual case. The thought occurs to us, however, that some risk may be run of setting up an ulcerative action in the

vessel walls by the prolonged application of a loose ligature, upon which must be exercised an intermittent pressure, by the pulsation of the vessel itself.

ARTIFICIAL AID IN THE FORMATION OF CALLUS.

Prof. Helferich (*Archiv. f. Klinische Chirurgie*, Band 36, 4. Heft, 1888). In cases of delayed union, and even in normal cases, to hasten the natural process of repair, H. advises the application of an elastic rubber bandage in such a manner as to retard the return flow of venous blood, by this means favoring an increased amount of pabulum to the field of repair, thus indirectly augmenting the formation of callus. The patient must be taught to regulate the pressure, attention being directed to the condition of the nails, in order that the bandage may be adjusted to suit the varying condition of congestion present. Œdema may be controlled by the application of a flannel bandage to that portion of the limb below the site of fracture. It is claimed that by this method the cure, in normal cases even, is considerably shortened. The process of repair is hastened by keeping the limb in a dependent position. The presence of small erosions at the site of fracture is not a contra-indication to the use of the elastic band. The time of application is of some importance, a too early application leading to too active hyperæmia; while, on the other hand, if too long delayed, the period of time in which the action will take place has passed. In wired compound fractures and in resections, pressure may be applied in from five to fourteen days after the operation, providing inflammatory symptoms are absent.

Thomas, of Liverpool, has recommended a procedure, which is known as the percussion method, to hasten the repair in delayed bony union, in cases of imperfect union, and in ununited fractures. This consists of percussing, once in a day or two, the site of the injury with a small copper hammer for five minutes or more, and subsequently bandaging the parts firmly.

It is suggested that the formation of varicose veins may be an objection to the method of Helferich. Further, it is quite clear that the method is not to be thought of in tuberculous subjects, as well as in cases of large open wounds at the site of fracture, or where a gap is left by resection of bone, removal of tumors, etc. The question of its applicability to atrophic members is an open one.

COMMUNICATION OF TUBERCULOSIS BY RITUAL CIRCUMCISION.

F. S. Eve (*The Lancet*, January 28th, 1888), relates the case of a Jewish child, in whom, six weeks following the usual rite, a small swelling appeared in each groin. They were found to be filled with caseous material, which, upon being inoculated beneath the skin of guinea pigs, gave rise to tuberculosis in the latter. The "Mohl," or person performing the rite, had ejected some wine from his mouth over the cut surfaces of the prepuce. It was subsequently learned that this person had died of pulmonary consumption shortly afterwards. Another child in the same house, operated upon by the same person, suffered from the same infection. Both children finally recovered.

Similar experiences have been recorded by Eisenberg (Berlin Med. Woch., No. 35, 1886), and Meyer (Centralblatt f. Chirurgie, No. 46, 1887). Of greater importance, because of a probably greater frequency, is the transmission of syphilis in this manner. A group of cases of this kind were recently collected and made the subject of study at the London Hospital.

TRANSPLANTATION OF THE SKIN.

Baratoux and Dubonsquet (Progres. Med., No. 15, 1887). D. treated two extensive wounds caused by burning, in which no attempt at cicatrization seems to have been made, although granulation was progressing well, by transplantation. Simultaneous auto-transplantation, and pieces of skin from a frog's back the size of a thumb-nail, was practiced. Most of the latter lost their pigmented aspect after ten days, and adhered well, taking on the natural color of the human skin. The wound where the frog's-skin transplantation had been performed healed more rapidly than the other where human skin was used, the cicatrix being softer as well. B. treated cases of ulceration of the nose, and also perforations of the membrana tympani, successfully by transplantation of frog's skin, healing taking place in from one to two weeks.

In three old cases of perforation of the drum membrane, the margins were freshened by touching them with nitrate of silver, and a piece of frog's skin attached. In three days a cicatrix had formed, with considerable

improvement in the hearing. Transplantation must be practiced with a healthy granulating wound, hæmorrhage being avoided. According to the authors, the wound should be irrigated with a strong solution of carbolic acid, and dried; the pieces of skin should also be washed in a weak solution of carbolic acid.

It suggests itself to the abstractor that still better results would be obtained by substituting a sterilized solution of chloride of sodium, say of the strength of 6 to 1,000, for the strong carbolic solution recommended, to be used just prior to the operation. The changes produced in the vessels and their contained blood by the use of strong disinfecting solutions are calculated to prevent early adhesion of the new skin. At least such is the general experience of recent observers. Reliable disinfection of the ulcerated surface may be obtained by keeping the parts covered with gauze wrung out of a 1 to 12,000 solution of potassio-mercuric iodide for a day or two previously.

OBSTETRICS.

BY CHARLES JEWETT, A.M., M.D.,

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THE IMPROVED CÆSAREAN SECTION.

Garrigues (Am. J. M. S., May, 1888,) describes in detail a successful case of Cæsarean section with observations on the *technique* of the improved operation. He prefers a long abdominal incision, and eventration of the uterus before opening it, the advantage claimed being the easier application of the rubber constrictor. The constrictor is more manageable if held in the hand of the assistant instead of the clamp, since it can be loosened and tightened as required. To prevent prolapse of intestines he sutures the upper end of the abdominal incision before turning out the uterus, tying the sutures before that organ is opened. Extraction of the fœtus by the head is much easier than by the feet. When a long uterine incision is required, it is better to go an inch into the fundus than to extend the wound into the lower segment, which may cause troublesome hæmorrhage.

Removal of the ovaries for the prevention of subsequent pregnancies he thinks not justified. The omentum he pushes up above the uterus to prevent adhesions to the suture line and the consequent danger of subsequent intestinal obstruction.

Dr. Garrigues believes Cæsarean section safer than *difficult* extraction through the natural passages.

Eleven Cæsarean sections were done in this country between December 16, 1886, and February 24, 1888, (Dr. R. P. Harris) all by the improved method but one. Six women and eight children were saved. Six of the operations were performed in hospitals, saving five women; five in private practice, saving only one. All the five hospital cases operated by the improved technique were successful. The bad results in private practice Dr.

G. ascribes to imperfect antisepsis. He alludes to the tardy adoption by our own countrymen of the antiseptic methods in general obstetric practice which have met with almost universal acceptance elsewhere—in Germany midwives being even compelled by law to use antiseptic precautions in every case of confinement.

Including the above-mentioned case, one hundred and sixty-three Cæsarean sections had thus far been done in the United States (Harris). One hundred and seventy to date of this writing.—ED.

The paper concludes with a detailed statement of the *modus operandi* and after-treatment in the modern Cæsarean operation. (A loop of the constrictor can usually be readily passed over the fundus and slipped down to the cervix while the uterus is still in the abdomen as we have shown.) (A Case of Cæsarean Section, N. Y. M. J., August 29, 1885.) Traction upon the constrictor perfectly occludes the short abdominal wound during the incision of the uterus, eventration taking place as the uterus collapses on removing the fœtus. The advantage, therefore, of extending the abdominal incision some inches above the umbilicus in all cases and turning out the uterus before opening it may be doubted. It is sometimes, however, impossible or difficult to apply the constrictor to the uterus in situ. Extraction of the fœtus by the head is certainly easier than delivery by the feet as advised by most writers.

The comparative results of induced labor, version, perforation and Cæsarean section in the Dresden Clinic have been recently considered in a series of papers by Leopold and his assistants, Korn, Lohman and Praeger.

The maternal mortality was as follows:

Induced premature labor	2.2 per cent.
Version and extraction	4–8 per cent.
Perforation	2–8 per cent.
Cæsarean section	8–6 per cent.

The fœtal mortality was in

Premature labor	33.4 per cent.
Version and extraction	41. per cent.
Cæsarean section	13. per cent.

Leopold concludes that while Cæsarean section cannot yet be substituted in all cases for craniotomy, it is at least justifiable as an alternative when the following conditions are present or possible, viz.:

Complete asepsis.

The patient strong and not long in labor.

The foetal heart-beats normal in rate and rhythm.

Strauch (Arch. f. Gyn.), analyzing the results in twenty-eight cases of induced premature labor arrives at like conclusions. While the mortality in cases of the mothers was *nil*, the foetal mortality was fifty-five per cent. The rights of the foetus demand a more frequent choice of Cæsarean section, the mortality of which by the modern method is thus far 11.8 per cent. for the mothers and 8 per cent. for the children.

Dr. Felice La Torre, of Paris, reaches the conclusion from extensive clinical research that craniotomy or premature labor is better than Cæsarean section, since the former saves all the mothers.

Krassowski (Arch. f. Gyn., B. 32 H. 2) reports five Porro and two Saenger operations saving six mothers and five children. K. uses thymol 1:1,000 for instruments, and the biniodide of mercury 1:4,000 for the hands, etc. He seals the abdominal wound with collodion to which biniodide of mercury has been added.

Zweifel reports six additional cases of the Cæsarean operation after Saenger, saving five mothers and all the children.

THE MECHANISM OF LABOR IN HEAD PRESENTATION.

Sutugin (Sammlung Klin. Vorträge, No. 310) makes an important contribution to the knowledge of this subject. The paper deals with an "almost wholly neglected factor in the mechanism of labor," namely, the position of the foetal trunk in utero, the mechanism of the trunk movements and their effect upon the positions of the head during labor. He first shows that the views commonly accepted with reference to the position of the foetal trunk during pregnancy are in part erroneous. Observations by the author in six hundred and sixty cases, published in 1875, established the fact that before labor, in either right or left positions, the dorsum of the child is almost invariably turned to the mother's back, the vertebral column of foetus, as a rule, lying but little to one side or the other of the spinal column of the mother; and, furthermore, the changes of position during pregnancy, as from right to left, probably take place by rotation along the

posterior wall of the uterus. On the occurrence of energetic uterine contractions, especially at the beginning of labor, the back of the child is sometimes rotated to the mother's side. Kehrer has confirmed the conclusions of Sutugin in observations upon certain of the lower animals. It may be noted in passing that, according to Kehrer's observations, gravity is a subordinate factor in determining the attitude of the fœtus in utero.

The author of the paper declares that, in a large number of cases examined during the last twelve years he has not in a single instance found the back of the child turned wholly forward during pregnancy, not even in first positions of the head. He has more recently made a study of the varying positions of the trunk during labor. Early in the labor, in first positions of the head, the trunk rotates so that the back of the child looks sideways, the shoulders lying in a plane parallel with the introitus. The breech rotates more slowly than the shoulders, the spinal column of the child thus assuming the form of a spiral during delivery. The fœtus, therefore, in its descent moves in a screw-like direction around its own axis, but the back of the child is not turned forward even during parturition, as authors generally have assumed. These views are born out by the frozen sections of Chiara, Waldeyer, and Shroeder. The rotation of the head is in part due to the rotation of the trunk, "the torsion of the axis of the fœtus," and is not to be referred solely to the action of the pelvic planes. The author claims that a torsion of the uterus upon its axis similar to that of the child also occurs. With reference to the etiology of the torsion in case of uterus or fœtus, he ventures no explanation.

MANAGEMENT OF THE PLACENTAL STAGE OF LABOR.

Fehling (Sammlung Klin. Vorträge, No. 308) compares the views and practice of authorities in the treatment of the third stage of labor. The various methods of placental delivery that have been advocated by different writers are recounted. The reaction against Credé in favor of expectancy, first started by Dohrn and Ahlfeld, has resulted in proving the inferiority of the latter plan, and in a return to more active methods. In a large number of German clinics, the uterus is allowed to rest immediately after the expulsion of the child, without friction. When the placenta lies detached in the lower uterine segment, which is generally the case after fifteen or twenty minutes,

nothing is to be gained by longer waiting. The author is favorably disposed toward the practice of Credé, which as he says, has never been shown by its adversaries to be capable of harm when properly conducted. With reference to the mechanism of placental separation, both theoretical considerations and clinical observations favor the views of Duncan, yet the question is not settled. Retained membranes may be removed with the aseptic hand. Yet Credé and Olshausen consider the retention of even the whole chorion free from danger, and clinical experience has shown better results by the expectant plan in case of retained membranes than by interference. These results, the author thinks, in the light of Döderlein's researches, are explained by the fact that the hand may transport the peccant germs from the vagina into the uterus. The active plan, with a preliminary vaginal disinfection and a vigorous asepsis throughout, should yield better results than expectation. Interference with these precautions is, at least, justified in case of atony and hæmorrhage or fever, including, if need be, the use of the curette and subsequently ergot. Dührssen's method of tamponing the uterus in post-partum hæmorrhage with iodoform gauze is favorably mentioned.

In the event of cervical tears causing troublesome hæmorrhage, Kaltenbach, Schroeder and Leopold practice immediate suture. The author thinks the vaginal tamponade is generally to be preferred. Yet, in certain cases the suture may also be required, or the application of the perchloride of iron on cotton pledgets to the bleeding surfaces.

Credé (Arch. f. Gyn., B. 32, H. 1) again discusses and defends his method of managing the placental delivery which he prefers to call the external method.

The duration of the third stage need not in the majority of normal labors exceed fifteen to thirty minutes. In many instances a more expectant plan of treatment is better. In occasional cases more rapid delivery is demanded in the interest of the mother. Since the method is free from danger when properly conducted, the expulsion of the placenta may be hastened within reasonable limits if for no other purpose than to save the time of the attendants and to spare the sufferings of the patient. He claims that the amount of blood-loss is diminished under his treatment of the placental stage, and that the membranes are not more frequently retained. Furthermore, he believes the retention of portions of the membranes or placenta to be harmless in an aseptic condition of the passages.

The method of Credé, briefly restated, is as follows: First apply gentle, painless friction in a circular direction over the anterior wall of the uterus, laying the hand flat upon the abdomen. Bring the axis of the uterus in conformity with the axis of the pelvic inlet. If the placenta is not expelled after three or four pains assist the next contraction, at its acme only, by compressing the upper segment, grasping the fundus with the thumb in front and the fingers behind, at the same time using gentle downward pressure. Use slight friction only but no compression during the intervals between the pains nor even during the contraction except at its height. Success usually attends the eighth or tenth pain.

SCARLET FEVER: ITS RELATION TO PUERPERAL FEVER

Boxall (Br. Gyn. J.) in sixteen cases of scarlet fever in childbed found septic manifestations in but one. In forty lying-in patients exposed to the scarlatinal poison the puerperium was entirely normal. Three hundred patients or more were admitted to the hospital during the epidemic of scarlatina therein, yet a comparison of the morbidity during this time with that which immediately preceded the outbreak showed that the prevalence of scarlet fever in the hospital exerted no appreciable effect upon the course of the puerperium in patients who escaped scarlatina.

Galabin (Br. M. J.) thinks there is strong evidence of the bacterial relation of puerperal sepsis to scarlet fever. Septicæmia does not represent a distinct entity like scarlatina. Cheyne found the common microbes of suppuration in the blood of scarlet fever patients not infrequently.

MITRAL STENOSIS AND THE THIRD STAGE OF LABOR.

Dr. D. B. Hart (E. M. J., Feb. 1888,) reports eight cases of this complication with seven deaths. With reference to the etiology Dr. Hart thinks the progress of the cardiac lesion is greatly accelerated by the increased amount of work imposed upon the crippled heart during pregnancy. At the beginning of labor, therefore, we may get failure of compensation, dilated heart and engorgement of the lungs. At the close of the labor, if free hæmorrhage does not occur, the extra blood before

accommodated in the utero-placental sinuses is returned to the right heart. Death is therefore liable to occur in the third stage from over distention of the right heart.

Dr. Ballantyne (E. M. J., March, 1888,) adds two more cases to the above record, both terminating fatally. Sphygmograms obtained in these cases show that the period immediately following the expulsion of the placenta is the one of greatest danger, and they are entirely consistent with Hart's theory of the cause of death.

TREATMENT OF ABORTION.

Fry (Am. Obs. J., June, 1888,) advocates the use of the galvanic current as a substitute for the curette for the removal of retained fragments of the secundines. He uses a current of sixty to ninety milliamperes with the anode in the uterus. The application is continued from six to ten minutes and repeated on alternate days. The placental tissue, owing to its relatively low vitality, is destroyed without injury to the uterus itself. Separation and expulsion follow. Hæmorrhage is relieved by the well known hæmostatic action of the positive electrode. Dr. Fry also claims antiseptic properties for the positive pole since here are liberated oxygen and chlorine in a nascent state and also acids.

Goodell thinks the curette an inefficient instrument for the evacuation of the uterine cavity and liable to injure the uterus. He advocates polypus forceps. Parrish finds the curette deceptive. He uses the finger. Longaker prefers the finger. [A Sims' speculum, a dull curette and a strong, straight uterine dressing forceps, with its joint two and a half inches from its distal end need never fail. The finger is awkward, difficult, painful, and sometimes requires preliminary dilation of the cervix. It cannot, moreover, be so easily sterilized, and even though clean primarily is liable to carry septic organisms from the vagina. Injuries to the uterus are for the most part the fault of imperfect asepsis.]

PLACENTA PRÆVIA.

Obermann (Arch. f. Gyn., B. 32 H. 1.) discusses the treatment of placenta prævia by version with the results obtained in sixty-four cases at the Leipsic clinic. The method, which has become known as Hofmeier's method, he states as follows:

Perform bimanual version with deliberate extraction in case of much hæmorrhage. The podalic extremity of the child makes an effectual tampon. Massage of the uterus during extraction is advised to aid expulsion. Iodoform gauze tampon may be used in case of hæmorrhage in the early months. The colpeurynter is recommended in case of hæmorrhage with a closed cervix. Alcoholic stimulants are given early and often.

The results in the sixty-four cases were eighty-nine per cent. of the mothers and forty-seven per cent. of the children saved.

Nordmann, of Dresden, condemns early resort to operative procedures as a routine measure, a more expectant plan of treatment being competent in a certain proportion of cases.

Robt. Barnes (Br. Med. Jour., March 3d, 1887) sums up his views substantially as follows: The hæmorrhage in placenta prævia proceeds from so much of the lower zone of the uterus as is laid bare by separation of the placenta during canalization. This comprises all that portion of the uterus that lies below the equator of the foetal head. When canalization is complete the hæmorrhage is almost invariably arrested spontaneously by retraction of the lower zone thus freed. Until canalization is completed flooding is liable to persist, but after that process is accomplished the case becomes practically a natural labor. The too prevalent idea that the hæmorrhage is unavoidable and must go on till delivery is erroneous and mischievous. Enough placental attachment usually remains after complete dilatation to preserve the life of the child. The foetal life is not necessarily compromised except in certain extreme cases of complete central placenta prævia. His views of treatment follow as a corollary. Expedite the first stage, avoiding violence or precipitation. His caoutchouc bags accomplish this indication and control hæmorrhage. Detach the placenta with the finger from the zone below the equator of the head, thus permitting retraction and arrest of flooding. Rupture of the membranes and the use of the binder meet the indications in certain cases. The vaginal plug may be used in occasional instances if carefully watched. The os uteri moderately expanded and the placenta separated from the lower zone, hæmorrhage having ceased, wait.

With sufficient dilatation, deliver, if necessary, by forceps, version, or craniotomy. Dr. Murphy's success by this plan has been unexampled. [Dr. Barnes does injustice to version in placenta prævia, since he appears to assume that immediate extraction and violence to both mother and child are a necessary part of the procedure. The success of that plan he attributes in great part to the fact that, in carrying out the operation of turning, the placenta had probably been detached from the lower zone. These criticisms certainly cannot apply in case of external or bipolar version. With reference to Barnes' bags, it is safe to say, "the German teachers" are not the only practitioners who have found them, in many cases, more or less impracticable.]

OBSTETRIC SEPSIS AND ANTISEPSIS.

Auvard, writing to the *Annals of Gynec.*, April, 1888, says, while before the days of antiseptics it was better for a woman to be delivered in the street than in a hospital, the hospital ward is now less dangerous than the isolated lying-in chamber of the out-patient obstetric service. He points out the importance of improving the resisting power by use of tonics before the labor, in debilitated patients. In Auvard's practice every woman takes a thorough bath at the beginning of labor. The whole vulvar and vaginal surfaces and cervical canal are sterilized before expulsion begins, before obstetric wounds are developed. Asepsis before and during the birth is more effectual than the use of antiseptics at the close of labor and renders the latter unnecessary.

Sublimate soap and the sublimate solution, with which this soap makes a lather, are well rubbed into the surfaces to be cleansed by aid of the fingers. The dangers of sublimate poisoning do not obtain while the surfaces are intact.

In the event of septic developments during the puerperal period he thinks sublimate irrigation insufficient for genital antisepsis. The vulva and vagina should be scrubbed by aid of the fingers with the sublimate soap and solution. The uterus should be scraped with the curette. Auvard has devised for this purpose a curette with a hollow stem through which a stream of the antiseptic solution is kept flowing during its use. [The mercurial salts may be replaced with a 1:1000 hydronaphthol solution for use within the

passages after labor, though the danger from the former antiseptic may be greatly diminished by washing away the mercurial with a final injection of plain boiled water.]

AUTO-INFECTION IN CHILDBED.

Ahlfeld (Cent. f. Gyn., No. 52,) shows that it is not always safe to presume upon the impossibility of self-infection, and reports two cases in point. In rare cases infection may arise from organisms primarily present in the genitalia. [This goes to fortify Auvard's position.]

OCCURRENCE OF GERMS IN THE DISCHARGES FROM THE UTERUS AND VAGINA DURING THE PUERPERAL PERIOD.

Döderlein (Arch. f. Gyn., B. xxxi., H. 3,) finds in a series of carefully conducted observations, that in normal cases the uterine discharges contain no germs, while in the same patients numerous varieties of germ life abound in the vagina. Pathogenic organisms may occur in the vagina apart from any internal examination. These germs may gain access to the uterus of themselves when not carried by intra-uterine instrumentation or manipulation. These conclusions are confirmed by Kaltenbach.

The uterine lochia of women suffering from puerperal sepsis in any form invariably contain germs, the streptococcus pyogenes being constantly present.

SUBLIMATE SOLUTION.

Laplace has shown the importance of acidulating sublimate solutions for general antiseptic use. Neutral solutions on mingling with blood or other albuminous fluids become more or less inert by the precipitation of the albuminate of mercury. That precipitate is not formed in the presence of hydrochloric or tartaric acids. The proportion for sterilizing wounds should be five parts of the acid to one of the sublimate in one thousand of water. [Biniodide solutions require no acid. This is one of the many advantages of

the mercuric iodide over the bichloride for antiseptic use. In a series of experiments made for the purpose of determining the reaction of biniodide solutions on albuminous fluids we found that neutral solutions of the biniodide of mercury yield no precipitate with albuminous fluids. No reaction was obtained with a biniodide solution acidulated with hydrochloric acid in the proportion of five parts to the thousand. The addition of organic acids, such as acetic, citric or tartaric, as is well known, causes a precipitate of albuminate of mercury. There is no chemical incompatibility between sublimate soap and biniodide solutions.]

TRICHLORIDE OF IODINE.

One of the new antiseptic agents is iodine trichloride. In contact with organic bodies it gives off iodine and chlorine in the nascent state. The final products, moreover, hydrochloric and iodic acids, are well known oxidizing agents. The strength of solution used is one part in 1,000 or one in 500 parts of water. Such a solution is equivalent in sterilizing power to a 1:1,000 or 2,000 sublimate solution. [One disadvantage of this antiseptic is the powerful corrosive action of nascent chlorine and iodine upon metal instruments.]

DANGERS OF ANTISEPTICS.

Senger (Br. M. J., May 19, 1888,) has proved by experiments on dogs and rabbits that the antiseptic agents commonly employed are liable to cause degeneration of the kidneys. He injected into perfectly healthy animals corrosive sublimate, carbolic acid, etc., in one-twelfth the quantity necessary to kill them. Then on extirpation of one kidney he found in all cases, on microscopical examination, glomerulo-nephritis. He also found fatty degeneration of the liver, spleen, the heart-muscle, etc. The various antiseptic agents were found to be injurious in different degrees, corrosive sublimate being the most dangerous, then the others in the following order: iodoform, carbolic acid, salicylic acid, boric acid. These observations especially enforce the importance of avoiding the use of antiseptics in the abdominal cavity, or in other large cavities under conditions favorable to

absorption. Sterilized water or a five per cent. chloride of sodium solution should be substituted for use in the peritoneum. Senger has shown that the salt solution in no way injures the organs, and that it possesses moderate antiseptic power, killing the streptococcus pyogenes aureus in twenty-eight minutes.

EFFECT OF ERGOT ON THE INVOLUTION OF THE UTERUS.

Drs. G. G. Herman and C. O. Fowler (Br. M. J., Feb. 11, 1888,) discuss this question, basing their conclusions on the results noted in fifty-eight cases treated with ergot for a fortnight after delivery, and sixty-eight in which a single dose only of ergot was given at the close of labor. In the cases continuously ergotized the diminution of the uterus in size was more rapid than in those who received but a single dose. On the duration of the lochial discharge no appreciable effect was observed from the use of ergot.

Dr. Boxall has made similar observations on two parallel series of cases of one hundred each. Every alternate patient was given a mixture containing ext. ergot. ammon., ℥ xv., t.i.d., during the first three days after labor. Dr. B. concludes from the experience of these cases that the practice of giving ergot as described tends to prevent the formation of clots, to hasten their expulsion, and to diminish the frequency, intensity and duration of after-pains. That if omitted at first and given afterwards ergot tends to promote the expulsion of clots and to relieve after-pains.

ANÆSTHESIA WITH CHLOROFORM AND OXYGEN.

Dr. Kreutzmann (Cent. f. Gyn.) recommends a mixture of oxygen and chloroform vapors as an anæsthetic in obstetric and surgical practice. The mixture may be made by passing freshly prepared and pure oxygen through chloroform on its way to the inhaler. Neudörfer injects a small quantity of chloroform into a balloon filled with oxygen, administering through a face piece. It is claimed for this method, that anæsthesia is at once established after a few deep inspirations without the least excitement, and that there are no disagreeable after-effects, the patient awaking promptly on ceasing the anæsthetic as from a refreshing sleep.

Veit, of Berlin, has operated in seven cases of tubal pregnancy before rupture, in all successfully.

Breisky, of Vienna, has recently performed the first successful laparotomy for the removal of an ectopic viable fœtus in which all the fœtal appendages were at the same time removed. The operation was done at the end of the eighth month. Breisky advocates primary laparotomy with extirpation of the entire sac in preference to the secondary operation.

Brunniche (Cent. f. Gyn.) treated a case of vomiting of pregnancy successfully by feeding the patient through a tube introduced into the upper part of the œsophagus.

Duncan (London Lancet) reports three cases of hyperemesis in which complete relief followed the application of cocaine in five per cent. solution to the vaginal vault and cervix.

Saenger says the process of uterine involution is not a fatty degeneration but normal metabolism.

Dr. Temple, of Toronto, Canada, in a case of post-partum hæmorrhage, hot water and other measures having failed, injected the uterus with a tumblerful of undiluted brandy. Prompt contractions and arrest of hæmorrhage followed.

QUININE AS AN OXYTOCIC.

Dr. Cordes finds in quinine an efficient oxytocic in incomplete abortion. He administers two and a half grains hourly till the desired result is accomplished.

REVIEWS.

THE APPLIED ANATOMY OF THE NERVOUS SYSTEM—Being a study of this portion of the human body from a standpoint of its general interest and practical utility. Designed for use as a text-book and as a work of reference. By Ambrose L. Ranney, A.M., M.D. Second edition, revised and enlarged. 8vo. Profusely illustrated. Cloth, \$5.00; sheep, \$6.00.

New York: D. Appleton & Co., 1, 3, and 5 Bond Street.

A second edition of this work has just been issued and, as the author says, has been enlarged. It contains 791 pages including the index; and is divided into four main parts, the first part treating of the brain; 2d, the cranial nerves; 3d, the spinal cord; 4th, the spinal nerves.

It is very difficult to give a review of a work like this, which treats so largely of the anatomy of the nervous system; and which is the most difficult part of anatomy, and a great deal of which is not positively settled, but is under sharp discussion by those who pay special attention to the subject.

The work is a more or less successful compilation, which is prepared with the idea of its being used as a text-book. Its size indicates that it contains an enormous quantity of material on the gross and fine anatomy, but excursions are also made into the domain of medicine to illustrate the application of the anatomical knowledge to the explanation of symptoms in disease; and quite frequently physiology and pathology are dealt with.

As a text-book the work appears to me a great deal too large; and the treatment of the subjects too diffuse; and often not clear; this is especially so in the anatomical part. Cerebral anatomy is one of those subjects of which it is very difficult to treat in a clear and comprehensive manner so that others can understand it; and for this reason a text-book should be small

and contain only such anatomy as is clearly made out and can be made practical use of as applied anatomy. The finer anatomy and the study of the course of fibres, etc., should be taken up as a special work, and studied with patience on specimens, sections, etc.

A great many digressions into physiological questions might have been left out, and some other subjects properly belonging to general medicine, which the student could best study in some of the recent works on diseases of the nervous system, or at the clinics. As an illustration of what is meant, take, for instance, a consideration of the tendon reflex on page 576, and the short imperfect sketch on page 621, on progressive muscular atrophy, when on the next page is a figure of a man, forty-five years old, with progressive muscular atrophy, from Freidreich's work; a case which, by the way, is probably not a case of chronic myelitis of the anterior horns, but one of the cystrophies. The subject of progressive muscular atrophies is now undergoing close study, and a large number of cases are not dependent upon lesions of the anterior horns, but are due to changes in the muscles themselves. The work of Freidreich, Erb, Lichtheim, Ladaur, Charcot, Landonzy and Dejeuni, and many others, have placed the subject in a different light from that in which it was viewed some ten years ago.

Subjects like these can be found more satisfactorily treated in other works, and are altogether out of place in a book like this and only adds to its bulk.

The descriptions are often such as to confuse and mislead a student; for instance, take the opening chapter on the brain, where the author says: "In man and the vertebrates, the cerebro and spinal axis may be divided into three separate portions, each perfectly independent of one another and yet very intimately connected."

Now this division is quite artificial, and is only for purposes of description, and these parts are not *perfectly independent of one another*.

Again, he says: "The nervous system of all animals may be subdivided into two distinct histologic elements, nerve cells and nerve fibres." What has become of the neuroglia and neuroglia cells; are they not important histologic elements in the nervous system of all animals? Without this supporting framework what would become of the nerve tubes and ganglion cells? And in many of the diseases of the central nervous system this

neuroglia takes on a very active condition, as is seen in such a disease as disseminated cerebro and spinal sclerosis.

On page 45, in speaking of the study of sections of the cortex, the author says: "By a judicious employment of gradually increasing powers in the microscopic objectives used, the general arrangement of the elements may be first mastered, and later on the minute details of each of the component parts may be studied."

This sounds like a most formidable and delicate task in the *judicious employment* of objectives in increasing powers in the study of these sections. No one should attempt to study the histology of the nervous system without previously knowing something about the use of the microscope and having some practical knowledge of general histology and pathology; to such a person the study of sections does not depend upon *the judicious employment of gradually increasing powers in the microscope objectives*; if he use a No. 2 and a No. 7 of Verick, or some objective of about the same magnifying power, it is then simply a question of studying the specimens with those powers and learning to see and understand what he sees; there is no mystery about it.

I will refer to one other paragraph on page 56, where it is stated:

"The central gray matter of the spinal cord. This has no connection with the higher senses. It is capable in itself of the simplest kind of reflex acts by means of the spinal nerves. These can be produced at the will of the experimenter, in the beheaded frog, when an irritation of the skin by an acid, etc., is created; and Robin has satisfactorily performed the same experiment upon a beheaded criminal. We have reason to believe that the spinal cord can be slowly and in a purely automatic way taught to perform certain series of muscular movements (as in playing scales upon a musical instrument, for example,) without any intervention of the higher ganglia."

This is physiology. Is it true that the central gray matter of the spinal cord has no connection with the higher senses? The complicated movements which are performed by a person playing on a musical instrument, like the piano, for instance, are more than a simple reflex action of the spinal cord; and we do not believe that the spinal cord can be taught to perform such movements without the intervention of the higher ganglia. When one is learning to play the piano or other musical instrument, the higher centres are constantly in action, guiding and regulating the muscular contractions

which go to make up the act of playing; after constant repetition under the guidance of the higher centres, the spinal cord and lower centres, as it were, learn and retain the combinations necessary to the performance of the act, all that is necessary is to start the particular combination, and the spinal cord will carry it on automatically.

The spinal cord cannot be taught to perform such complicated acts without the intervention and guidance of the higher centres to begin with.

Dr. Ranney has done a great deal of work in the preparation of this volume, and deserves much credit for his endeavors to collect this somewhat scattered material.

The work has numerous illustrations and diagrams, most of them exceedingly good, but we observe among them some of the worthless and often fanciful illustrations from Luys', which are reproduced here.

J. C. S.

MISCELLANEOUS.

DR. J. B. MATTISON.

Dr. Mattison recently spent several weeks in Bermuda, and the *British Med. Journal* in reporting a meeting of the British Med. Association, held in the Town Hall at Hamilton, says:

By request of the Society, Dr. J. B. Mattison, of Brooklyn, gave an address on the subject of narcotic inebriety. Attention was called to the extensive use of opium, chloral, and cocaine, notably in France, Germany and America. The genesis of the disease was a physical necessity in many cases. The speaker said in proper cases his plan—an original one—was to secure an entire narcotic disuse by regular reduction, in ten days, meantime bringing the nervous system under the sedative influence of bromide of sodium, in initial doses of thirty grains, at twelve-hour intervals, increasing the dose ten grains daily, and reaching, if required, a maximum of one hundred and twenty grains at the end of the withdrawal period. The resultant reflex irritation was treated by hot baths, cannabis indica, hyoscyamus, coca, and electricity, with a subsequent tonic regime. The prognosis was good as to recovery, but in most cases, sooner or later, there was a return to the narcotic, due to a renewal of the original cause, or to other conditions beyond control. A vote of thanks to Dr. Mattison closed the meeting.

Dr. Mattison is translating Erlenmeyer's *Die Morphiumsucht und ihre Behandlung*—the Morphia Disease and its Treatment; third and last German edition, the latest and largest work on the subject, which, with notes and comments by the translator, will be brought out the coming autumn.

LONG ISLAND COLLEGE HOSPITAL TRAINING SCHOOL FOR NURSES.

The graduating exercises of this training school took place on June 12th, at the hospital. Prof. Jarvis S. Wight presented the diplomas, and Dr. George G. Hopkins delivered the address. The following are the graduates of the class of 1888: Mrs. Elizabeth Raifstanger, Nellie E. Russell, Elizabeth Munday, Abigail Collins, Lucy Wood, Elizabeth Ritchie, Ellen Enright, Florence Jackson, Jennie E. Stuart, Minnie M. Flower, Florence Crompton, Signa Johnson, Eleanor Mary Senkler.

“POST TENEBRAS LUX.”

BY PROF. F. H. GERRISH, OF BOWDOIN COLLEGE.

The Era Prize Essay.—Reprint from the Pharmaceutical Era.

Originally every physician was his own apothecary, and at the present time probably a majority of medical practitioners dispense their own medicines, very rarely writing a prescription. These will have but a languid interest in the subject of this essay, which deeply concerns all physicians who are not their own apothecaries, and all compounding pharmacists.

In medical, as in every other science, the increase of knowledge so widened the field that it became impossible for one mind to grasp all the facts, and a division of labor took place, the part of the work which related to the collection, preservation and dispensing of drugs being assigned to a class of men who had displayed peculiar aptitude for that branch. Thus was constituted as a distinct occupation, the specialty of the apothecary, which, beginning as a department of medical science, is inherently honorable, and has been so developed that it gives scope for a lifetime of fascinating research, elevating study, and profitable endeavor, independently of any proper work of the modern physician. The two callings are, for the purposes of this discussion, as in their best actual operation, practically distinct; and yet they are not independent, but interdependent. The greater part of the physician's labor would be vainly spent, were it not supplemented by the service of the pharmacist; the latter's business would cease to have a reason for existence but for the vocation of the former.

In this paper it will be taken for granted that the physician is well educated and experienced in his profession, that the apothecary knows his business thoroughly, and that both are actuated by high moral purposes. The grievance of neither, therefore, will result from the intentional wrong-doing of the other, but from his thoughtlessness or conservative adherence to long-established custom. The honesty of each being presupposed, such a charge as the substitution of an inferior article for some ingredient in a prescription, or the false insinuation that a mistake in the medicine is due to

the compounder's carelessness need not be raised. Let us consider the grievances of each against the other.

The physician complains that the apothecary exceeds his function by prescribing for the sick. A person applies to the pharmacist for a remedy for a specified disease. The latter consults the dispensatory, finds a number of medicines mentioned in the therapeutical index under the name of that malady, selects one, and sells the article to the patient. He regards the protest of the doctor merely as the wail of one who is disappointed at not getting a fee for prescribing. The physician has a right to complain of those who prescribe for any but the indigent without a professional fee, for this makes it vastly harder for him to collect the charges to which he is entitled; but he has higher ground than this. With him the first step in every case of disease is diagnosis, without which prescribing is simply drawing a bow at a venture, with small probability of penetrating a joint of the harness; and he insists that neither the apothecary nor the patient is qualified to make a diagnosis. The determination of the character of a disease is not a simple matter, often baffling the profoundest learning and the broadest experience, and, in most cases, requiring special knowledge and discriminating judgment. The most obtrusive symptoms are by no means necessarily the most characteristic; a given symptom may be the accompaniment of different diseases, and sometimes attends pathological changes of diametrically opposite nature. But one who is uninstructed in this branch has nothing but symptoms to guide him, and therefore frequently, if not commonly, is led into error, which may produce the gravest results. The educated physician is the only person who is equipped to solve the problems of disease; and it is, in the long run, cheapest, even from the financial point of view, for one who is ill to obtain competent medical advice. Therefore, considering merely the welfare of the patient, the physician deplores the custom of counter-prescribing.

The universal practice among apothecaries of refilling prescriptions indefinitely, without the sanction of those who wrote them is frequently the subject of adverse criticism by physicians. "But," says the pharmacist, "is not the prescription mine?" Probably the reply will be affirmative, though this is a question not entirely settled in the minds of those concerned. Granting, however, for the sake of the argument, that the apothecary is the rightful owner of the prescription, he triumphantly asks, "May not one do as he chooses with his own?" Not always, by any means, unless he elects to

use his property in a way which will not imperil the welfare of others. The law puts many restrictions on the natural rights of ownership, for the purpose of shielding society from the ignorance, the malice, and the cupidity of proprietors. In the matter of prescriptions there are no such limitations; but in this, as in so many other affairs, we should be controlled not merely by considerations concerning our legal rights. The unwritten, but greater, law regards the question from a higher plane; and from this point of view one sees that there are moral reasons which should restrain the apothecary in these premises. Let us look at prescription-refilling in its remote as well as its immediate bearings.

The physician writes his prescription for one occasion and for one only. He designs it to fulfil the indications in a particular case at a given visit. It is often in the highest degree improbable that the conditions of the case will remain the same for a given length of time; and, when they change materially, another prescription is needed to satisfy the altered requirements. Of these things many patients and apparently, all apothecaries are accustomed to take no note; and, therefore, the prescription is refilled for the same person on many occasions, when something very different ought to be furnished, the expected improvement does not occur, and the physician is blamed for the failure. The prescription, intended for the treatment of a certain condition afforded relief and changed the state of affairs. Thereupon, further advice should have been sought by the patient, his physical condition investigated anew, and another recipe given by the physician, if he deemed it desirable. To hold the doctor responsible for the ill effect or lack of effect of his prescription in circumstances unlike those for which it was designed is obviously unjust. Yet this is done habitually, both in the case just supposed and in those other instances, quite as common, in which the recipe is compounded, not for whom it was written, but for some other person, who supposes (on what slender evidence it is appalling to contemplate), that his ailment is identical with that of the original patient. The refilling really amounts to the apothecary's abetting the patient in self-treatment. There is a trite remark to the effect that he who undertakes to be his own lawyer is sure of a fool for a client. In legal actions the matters at stake mostly relate to property; in medical affairs, health and life are involved. Can anything better be said of the wisdom of him who, without adequate training, undertakes to settle questions in which his very existence is concerned, than of the sagacity of the man who, with

no competent knowledge of law and the methods of the courts, presumes to manage a case in which merely a sum of money may be lost?

But undesirable as is the custom of pharmacists in this respect, it is unreasonable in physicians to grumble at it, until they reform a habit of their own, which encourages the practice which they deprecate. When a physician finds a sick man improving with a certain plan of treatment, and wishes the same medication continued, usually he simply tells the patient to have his bottle refilled at the apothecary's. The latter, knowing that this is the usage of the medical profession, is accustomed to suppose—constructively, at least—that, when the bottle is returned to be replenished, it is brought at the desire of the doctor; and he again compounds the medicine, as he would not be justified in doing, if the method of physicians was different. If the latter would habitually rewrite their prescriptions, or indite orders for repetition, whenever they wish a continuance of treatment, apothecaries would have the best possible sanction for supplying medicines to patients a second or a thousandth time, and would have no moral warrant for such action without the written direction of the authors of the recipes. When the doctors adopt the plan suggested, a violation of their request that their prescriptions shall not be repeated without their written order will furnish abundant ground for complaint; until such change of method is established, the consistency of their protest does not conspicuously appear. By the practice proposed, the apothecaries would lose some business, indeed, but the community would experience an immeasurable gain.

The sale of so-called patent medicines by pharmacists is a continual irritant to doctors. Their objections have the same basis as in the preceding case, namely, that apothecaries ought not to encourage the people to prescribe for themselves. A distinguished physician once said that drugs do more harm than good—a statement which will not be disputed by those most conversant with the facts. But this is no argument against the employment of medicinal remedies in any proper way. Drugs taken by advice of educated, competent physicians do an immense amount of good; the injury comes almost altogether from their administration independently of the recommendation of qualified medical men. Concocted, not to benefit humanity, but to enrich their manufacturers, advertised as positive cures of diseases which the utmost skill of the medical profession cannot control, sometimes inert, sometimes dangerous from the poisons which they contain, pressed upon the attention of the people with impudent persistence

and colossal mendacity, the infamous compounds, called patent medicines, are purchased by the credulous public in almost incredible amounts. Nobody knows as well as the pharmacist what quantities of drugs are sold in this form; nobody knows as fully as the physician the enormously baneful effects which they produce. If apothecaries would refuse to handle this class of goods, they would confer a wonderful blessing on humanity, by demonstrating their belief in the injuriousness of these articles, and thus bringing them into disrepute; and they would vastly dignify their profession by displaying its sense of moral obligation, even at the sacrifice of a lucrative part of its business.

Apothecaries complain that, to meet the demands of prescriptions, they are obliged, at great expense, to keep in stock a large number of whimsical preparations, and also many brands of the same standard medicines, when really one is practically as eligible as another. There is force in this complaint. As a rule, the former class of preparations has but an ephemeral popularity, and, as for the latter, there cannot often be need of specifying the goods of a particular manufacturer; for a worldly-wise, not to say honorable, pharmacist certainly may be trusted to furnish only fine articles, that he may keep the favor of those upon whose good-will his legitimate business depends. If a special brand is deemed necessary by the doctor, he can request a convenient apothecary to procure a quantity, and then can send to his shop the patients who require this article. Thoughtful consideration of this grievance will dictate some such course and effect its redress.

Occasionally, in places where apothecaries are accessible, physicians furnish their patients with medicines. This the pharmacist rightly regards with disfavor, as diminishing his legitimate business. The truly wise physician shuns this practice, perceiving that he cannot be as competent a pharmacist as is one who is nothing else, and furthermore, preferring to avoid the suspicion of administering medicines unnecessarily for the purpose of making a profit on them. Enlightened selfishness prompts him to encourage scientific pharmacy as a specialty.

BROOKLYN VITAL STATISTICS FOR JUNE, 1888.

By J. S. YOUNG, Dep. Commissioner of Health.

Data of comparison:

Population, estimated on July 1st, 1888	793,960
Inhabited houses, about	93,000

Average birth-rate per 1,000 for ten years (returns incomplete).

In the month of June, 1888, there were 1,513 deaths, the rate of mortality being 23.78 in every 1,000 of the population.

The number of births reported was	937
The number of marriages reported was	531
The number of still-births reported was	113

The mortality by classes and by certain of the more important diseases was as follows:

Causes:

1. Zymotic	434
2. Constitutional	251
3. Local	644
4. Developmental	121
5. Violence	63
Measles	6
Croup	8
Diphtheria	82
Scarlet Fever	45
Typhoid Fever	—
Whooping Cough	5
Malarial Diseases	4
Diarrhœal Diseases (all ages)	244

Diarrhœal Diseases (under 5 years)	233
Phthisis	137
Bronchitis	45
Pneumonia	80
All Respiratory Diseases	149
Bright's Diseases	35
Puerperal Diseases	17
Old Age	17
Suicide	10

Reported cases:

Diphtheria	214
Scarlet Fever	255
Measles	128
Typhoid Fever	—

During the month 13 cases of small-pox were reported, of which number 13 were confirmed as small-pox. 13 cases of small-pox were sent to hospital. No deaths from small-pox occurred in the city and 2 in the hospital.

Deaths by sex, color, and social condition were as follows:

Male	846	Female	667
White	—	Colored	21
Native	1102	Foreign	411
Married	328	Single	1006
Widows, Widowers, and not stated			179

Still-births (excluded from list of deaths) were as follows:

Males	60
Females	53
Total	113
Deaths in public institutions	115
Deaths in tenement houses	745
Inquest cases	156
Homicides	—
Suicides	10

Age Periods:

Deaths under 1 year	496
Deaths under 5 years	253
Total deaths under 5 years	749
Total deaths 5 to 20	126
Total deaths 20 to 40	223
Total deaths 40 to 60	233
Total deaths 60 and upwards	182

Certain foreign and American cities show the following death-rate for the month of June:

Brooklyn	23.78
New York	26.86
Philadelphia	18.98
Berlin	19.40
Dublin	23.58
Vienna	31.50
Paris	21.68
London	16.10
Glasgow	22.98

NEW INSTRUMENTS.

This is a bulletin of inventions and improvements of interest to the physician and surgeon, and is published quarterly by Leach and Greene, Boston, Mass. It is mailed free to any address.

ANTIFEBRINE.

Antifebrine as a proprietary name is controlled by patents, and when bought under this name it costs about thirty cents per ounce. The name acetanilide, for exactly the same substance, is not and cannot be controlled in any way, and under this name it can be bought for about fifteen cents per ounce. As acetanilide is about one-eighth the price of antipyrine, and effective in half the dose, it is far more economical to the patient.—
Pittsburgh Medical Review.

ENEMA LESIONS.

“Dr. Achilles Nordmann, of Basle,” says the *Lancet*, “has published a description of twenty-five bowel lesions due to the operation of administering enemata. They include three complete perforations and ulcers, and wounds of various depths and sizes. The causes of these lesions seem to have been the use of defective instruments, ignorance of the direction of the rectum, catching of the transverse fold on the tube, extreme irritation of the mucous membrane of the bowel, and obstructions caused by certain conditions of the uterus, the fœtal head, or an enlarged prostate. As a rule, these lesions are to be found on the anterior wall of the rectum, from one to seven centimetres from the anus. They are not always easy to diagnose, as other foreign bodies or caustics may produce similar appearances. Tubercular and hæmorrhoidal ulcers may be mistaken for them; these are, however, generally higher up. A perforating wound gives rise to paraproctitis, but this is not necessarily fatal, though a stricture generally results.”—*N. Y. Medical Journal*.



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